Cross-European coverage debt of frozen optical Earth-observation foundation models: a pre-deployment reliability report card and a conditional target-recalibration remedy

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Abstract

Frozen optical geospatial foundation models (GFMs) are increasingly deployed as off-the-shelf encoders for land-cover and local-climate-zone monitoring, yet practitioners lack guidance on which frozen encoder to trust after a regional move, or on how to restore a reliability guarantee once shift breaks it. We supply both through a distribution-free reliability audit (split-conformal, finite-sample guarantees) of frozen GFMs under real cross-European regional shift (thirteen optical Sentinel-2 encoders on GEO-Bench BigEarthNet-S2; five on EuroSAT-MS; plus a non-European multi-city LCZ extension on So2Sat LCZ42), packaged as a reusable pre-deployment reliability report-card protocol that returns four practitioner-facing numbers before deployment: the in-distribution false-negative-rate (FNR) guarantee, the geographic debt that guarantee incurs under shift, the labeled-data budget curve for repairing that debt, and the worst-class coverage a marginal guarantee alone would hide. Every frozen encoder meets its multi-label FNR risk-control guarantee in-distribution, yet all thirteen incur a real geographic debt under shift (FNR 2.6–4.7× nominal), bought back with larger sets, not a better encoder: FNR robustness rises with set size (Spearman $\rho = -0.66$, p=0.017), the paper's one adequately powered cross-encoder result. The textbook unlabeled fix does not close the debt—covariate-shift-weighted conformal sees its effective sample size collapse to 0.1–1.6% of the calibration set—so only a labeled target-region slice reliably restores the guarantee; target-label recalibration is the priced remedy,

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not a label-free method. Cross-encoder rankings remain exploratory and underpowered: a five-encoder pilot's stronger accuracy correlation regressed toward zero at n=13. The powered results are the 13/13 FNR debt and the worst-class gap; the audit is diagnostic, reusing existing conformal estimators, with preregistration outcome and exploratory-analysis disclosures stated in the main text.

Keywords: Earth observation, geospatial foundation models, conformal prediction, distribution shift, uncertainty quantification, reliability

1. Introduction

Geospatial foundation models (GFMs) are pretrained once and reused frozen as feature extractors for many downstream land-cover and climate-zone tasks. That reuse pattern is precisely why reliability under *geographic* shift matters: an agency that downloads Clay, Prithvi, or an SSL4EO checkpoint and fits a lightweight probe on a source region typically never retrains the backbone when the monitoring footprint moves west, south, or across a continental boundary. Accuracy-only transfer studies and leaderboard-style fine-tuning papers do not answer the operational question such deployments raise—whether a distribution-free coverage or risk-control guarantee attached to the frozen probe still holds after the move, and what labeled target budget is required if it does not. Beyond the encoders audited here (thirteen optical Sentinel-2 encoders on BigEarthNet, five on EuroSAT and So2Sat LCZ42), frozen GFM backbones now span single-sensor optical pretraining (SatMAE (Cong et al., 2022), Scale-MAE (Reed et al., 2023), SeCo (Mañas et al., 2021), RingMo (Sun et al., 2023), GFM (Mendieta et al., 2023), SatlasPretrain (Bastani et al., 2023)), multi-sensor and multi-modal pretraining (CROMA (Fuller et al., 2023), SkySense (Guo et al., 2024), SpectralGPT (Hong et al., 2024), AnySat (Astruc et al., 2025), TerraFM (Danish et al., 2025), Galileo (Tseng et al., 2025)), SoftCon-style contrastive EO pretraining (Wang et al., 2024), vision–language EO models such as RemoteCLIP (Liu et al., 2024), Landsat-family SSL (SSL4EO-L (Stewart et al., 2023)), hyperspectral-scale pretraining (SpectralEarth (Braham et al., 2024)), early HLS Prithvi releases (Jakubik et al., 2023), and pixel-timeseries pretraining (Presto (Tseng et al., 2023)). Recent surveys document how rapidly this literature has grown and how evaluation has remained dominated by task accuracy, few-shot transfer, and architectural coverage rather than deployment-facing uncertainty under spatial shift (Xiao et al., 2025; Lu et al., 2025; Huo et al., 2025; Yang et al., 2025); none of the families above is conformally audited under an explicit geographic train/deploy split, and our roster (Methods) is illustrative of the diagnostic rather than a leaderboard survey of this space. Foundation-model assessment has also broadened beyond accuracy: SustainFM frames geospatial foundation-model evaluation around transferability, generalization, efficiency, and deployment impact across sustainable-development tasks (Ghamisi et al., 2025). Reliability and robustness under related, but distinct, shift axes have begun to appear. SHRUG-FM (Gonzalez-Calabuig et al., 2025) evaluates SSL4EO ViT-S/16 (MoCo/MAE/DINO) checkpoints on burn-scar segmentation and reports expected calibration error (ECE 0.0248–0.0328) together with reliability-aware OOD flags, but it has no conformal-prediction arm, no spatial-holdout cross-validation of coverage debt, and does not evaluate Clay or Prithvi. GeoCrossBench (Tamazyan et al., 2025) instead stress-tests *cross-band / cross-sensor* generalization (train on one spectral configuration, evaluate with no band overlap or with additional bands)—an important operational failure mode that is orthogonal to the longitude-cut geographic shift studied here. EarthShift (Doerksen and Kerner, 2026) benchmarks effective robustness of GFMs across paired geographic, temporal, sensor, scale, and source shifts, while DSGR (Al-Emadi et al., 2025) isolates continent-scale spatial domain shift for land-use classification; both quantify accuracy-style OOD drops rather than conformal coverage debt on frozen probes. Explicit geographic or cross-city transfer in remote sensing is itself long-standing (Tuia et al., 2016; Zhu et al., 2017; Hong et al., 2023), as are parameter-efficient geospatial domain adaptation recipes for foundation backbones (Scheibenreif et al., 2024) and geographic area-of-applicability diagnostics for spatial predictors (Meyer and Pebesma, 2021). Geographic distribution shift is now also recognized as a structured instance of in-the-wild shift in the broader ML literature (Koh et al., 2021; Crasto and Rolf, 2025; Christie et al., 2018); those lines typically retrain, adapt, or location-condition the predictor, whereas we freeze the encoder and ask whether a conformal guarantee survives the regional move. Operational global land-cover products such as Dynamic World (Brown et al., 2022) and accessible satellite-ML pipelines (Rolf et al., 2021; Jean et al., 2016) likewise show that regional deployability—not only peak accuracy—is the binding constraint. Calibration itself is not entirely unstudied for deep classifiers: classic post-hoc calibration (Guo et al., 2017) and calibration diagnostics (Nixon et al., 2019; Mukhoti et al., 2020; Kumar et al., 2019; Minderer et al., 2021) remain the practitioner baseline, yet predictive uncertainty and ECE degrade systematically under dataset shift (Ovadia et al., 2019)—precisely the regime in which a geographic deploy can silently invalidate an in-distribution reliability reading. Uncertainty quantification for deep networks more broadly spans Bayesian, ensemble, and distribution-free approaches (Gawlikowski et al., 2023); conformal prediction is the distribution-free branch we build on (Vovk et al., 2005; Shafer and Vovk, 2008; Angelopoulos and Bates, 2023; Angelopoulos et al., 2021; Lei et al., 2018), including Mondrian / label-conditional variants (Vovk et al., 2003; Sadinle et al., 2019; Romano et al., 2020; Cauchois et al., 2021), conformalized quantile regression for continuous responses (Romano et al., 2019), and conformal risk control (CRC) for controlling losses beyond miscoverage (Bates et al., 2021; Angelopoulos et al., 2024). Conformal methods have entered Earth observation: Singh et al. (2024) bring split and (class-conditional) Mondrian conformal prediction to EO pipelines (Google Earth Engine), GeoConformal (Lou et al., 2025) adds geographically-weighted conformal intervals for *spatial regression*, Mao et al. (2024) develop model-free conformal prediction for spatially indexed data under local exchangeability, and Kakhani et al. (2024) apply conformal intervals to remote-sensing soil-carbon regression; none of these audits frozen foundation-model encoders, nor coverage loss under an explicit geographic train/deploy shift. Weighted conformal methods for covariate shift (Tibshirani et al., 2019) and adaptive / online conformal procedures for changing environments (Gibbs and Candès, 2021, 2024; Barber et al., 2023) are natural theoretical remedies, but the former require usable density-ratio estimates we do not assume here, and the latter target sequential settings rather than a one-shot regional move with a frozen GFM probe. Closest in spirit, Fillioux et al. (2024) ask whether frozen vision foundation models are good conformal predictors, but on in-distribution image classification with no geographic-shift audit; and Aolaritei et al. (2025) develop Lévy–Prokhorov theory for conformal robustness to distribution shift—complementary theoretical machinery that they do not instantiate on Earth observation. Audited Conformal Prediction (ACP) offers a complementary classification-level remedy for pretrained models under unknown target shift: it uses a small labeled target sample and an auxiliary audit model to retain marginal coverage while improving conditional or group-conditional coverage (Zhou et al., 2026). Our contribution is different in scope and object: we audit frozen optical Earth-observation encoders as deployed across European regions, quantify region-conditional coverage debt in a practitioner report card, and test labeled target-region recalibration as the repair. ACP is a genuine alternative remedy for classification-level conditional reliability, but it does not replace the geographic coverage-debt framing or the deployment-facing comparison across frozen EO encoders studied here. Our geographic-shift framing is also distinct from unsupervised domain adaptation for remote sensing classification (Tuia et al., 2016; Zhu et al., 2017; Hong et al., 2023), which typically retrains or adapts the feature extractor itself against the target domain; we instead ask whether a *frozen*, never-retrained encoder's conformal guarantee survives a regional shift, and how to repair it with only a labeled target-region calibration slice, not further training. Consequently, *conformal coverage* reliability under shift—whether a conformal predictor's guarantee on frozen classification GFMs such as Clay and Prithvi survives a move to a new region—has received far less attention than accuracy transfer. The gap relative to accuracy-only EO foundation-model papers is sharp: those works answer "which encoder scores highest after fine-tuning or probing," whereas a pre-deployment reliability audit must answer "does the guarantee travel, what debt does the move incur, and what labeled target budget restores it." We ask a falsifiable question: under a real cross-European optical shift, does a conformal audit of a frozen GFM tell a practitioner anything that shifted accuracy and calibration do not? We find that it does, in two regimes, and we characterize precisely where it does not.

Roadmap and protocol.. Our primary contribution is a reusable pre-deployment reliability report-card protocol (§4.3; overview in Fig. 1): apply one fixed audit to any frozen optical GFM and read off four practitioner-facing numbers before deployment—its in-distribution false-negative rate (FNR) guarantee, the geographic debt that guarantee incurs under shift, the labeled-data cost of repairing that debt with a target-region slice, and the worst-class coverage a marginal guarantee alone would hide (§4.3 assembles the card for a median-accuracy encoder). We introduce no new conformal estimator; the audited findings below are what make each report-card entry non-obvious and worth running the protocol for.

The experimental narrative follows a preregistered falsification-versus-remedy logic that is preserved honestly in Discussion: the Stage-2 preregistered *universal*-restoration gate was not met (Clay restored at only 1 of 3 risk levels), so the encoder-dependent target-region recalibration reported in Results is an exploratory reframe of that falsification, not a confirmatory universal fix (§4.4). The protocol's first evidence axis is a within-encoder reliability diagnostic that needs no cross-encoder statistics: under a conformal-risk-control (Angelopoulos et al., 2024) bound on the example-averaged FNR on GEO-Bench BigEarthNet-S2, *every one* of thirteen frozen encoders meets the FNR guarantee in-distribution yet incurs a real geographic debt under shift (0.130–0.234, 2.6–4.7× nominal), which a labeled target-region slice repairs (0.002–0.036)—a repair that is expected by construction, since the target-calibrated guarantee is a verification, not an independent discovery.

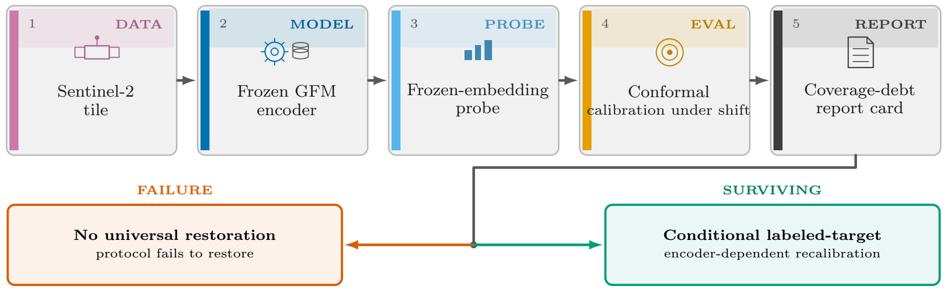

Figure 1: Overview of the pre-deployment reliability report-card protocol. Stages 1–5 (left to right), role-tagged DATA / MODEL / PROBE / EVAL / REPORT: (1) optical Sentinel-2 tile (EuroSAT-MS / GEO-Bench BigEarthNet-S2 / So2Sat LCZ42); (2) one frozen, interchangeable optical GFM encoder under audit on EuroSAT/BigEarthNet (Prithvi, Clay, SSL4EO-DINO, SSL4EO-MAE, DOFA; on So2Sat LCZ42 the ViT-S slot substitutes SSL4EO ViT-S/16 MAE for DINO—same five-encoder *size*, not an identical checkpoint roster); (3) frozen-embedding probe (logistic for single-label tasks, multi-label otherwise); (4) conformal calibration under real geographic region-transfer shift (source split-conformal vs. spatial-Mondrian target-region calibration; source = E/Central Europe, target = W/Iberian-Atlantic Europe), yielding per-class conformal sets; (5) coverage-debt report card with four practitioner-facing numbers—in-distribution FNR guarantee, geographic debt under shift, target-slice repair cost, and worst-class coverage a marginal guarantee alone would miss. Cross-sensor transfer (Sentinel-1) failed in-distribution control and is withheld. Fork from the report card: vermillion marks FAILURE (no universal restoration); bluish green marks SURVIVING (conditional, encoder-dependent labeled-target recalibration).

This debt is bought back with larger prediction sets, and that trade is the one adequately powered cross-encoder result in the paper: FNR robustness rises with prediction-set size (Spearman $\rho = -0.66$ at α=0.10, p=0.017; §3), an operating-point choice no accuracy scalar can name. The protocol's second evidence axis is the singleton-degeneracy boundary, a deterministic property rather than a correlation: the marginal single-label set collapses to a singleton above accuracy ≈ 0.80, so EuroSAT at full data is an explicit *negative control* bounding the audit's validity regime (So2Sat/BigEarthNet sit below it)—telling a practitioner exactly when a marginal audit of a frozen GFM is worth running at all. The protocol's third evidence axis is a remedy-failure result, and it is what fixes the report card's target-slice repair cost as necessary rather than optional: the textbook unlabeled fix for covariate shift does not close the debt. Covariate-shift-weighted conformal (Tibshirani et al., 2019) recovers only part of the coverage loss because its weights' effective sample size collapses to 0.1–1.6% of the calibration set—the density ratios are too heavy-tailed to be usable—so only a small labeled target-region slice reliably restores the guarantee; oracle and deployable label-shift reweighting (Lipton et al., 2018; Podkopaev and Ramdas, 2021) fail outright (worsening coverage for three of four encoders, ruling out a pure class-prior effect and pointing instead to a within-class conditional shift); and spatial-Mondrian conditioning adds nothing measurable over non-spatial target calibration, so the operative axis the protocol reports is access to labeled target-region data, not a new estimator (scope and estimator provenance are stated in §4.4). Only after these three do we report the weaker cross-encoder accuracy question, as directional context rather than a claim: accuracy is a weak and unreliable guide to *which* encoder is most robust—the least accurate encoder (Prithvi, mAP 0.432) is the most FNR-robust (0.130) and the most accurate is not—but the rank correlation over the full roster is only a weak, non-significant positive (Spearman $\rho = +0.32$, p=0.28; §3)—the strong +0.90 inversion visible in our first five encoders regresses toward zero when the roster is expanded, so we report it as directional evidence, not a proven inversion (§3). As the protocol's fourth evidence axis—a conditional and single-encoder finding that is exactly what the report card's hidden-worst-class-coverage entry is designed to catch—marginal coverage repair can hide a per-class collapse: on EuroSAT, after target-side recalibration restores the *marginal* guarantee, one land-cover class is still covered at only 0.850 under a 0.956 marginal—a failure invisible to accuracy and to ECE, reported as a single load-bearing existence proof, not a population result (§3). The protocol's fifth, secondary and descriptive evidence axis is an encoder coverage-debt ranking: the magnitude of geographic coverage debt ranks the frozen GFMs (Prithvi > Dynamic One-For-All (DOFA) (Xiong et al., 2024) > SSL4EO-DINO > SSL4EO-MAE > Clay)—though, because the source-split predictor is in the singleton regime, this ordering coincides with the shifted-accuracy ordering and is reported as descriptive, not as evidence the conformal layer beats accuracy. We establish these on a real EuroSAT-MS (Helber et al., 2019) cross-region split with five frozen GFMs (the original four plus DOFA (Xiong et al., 2024), added on the EuroSAT grid), corroborate them on GEO-Bench BigEarthNet-S2 (Lacoste et al., 2023; Sumbul et al., 2019) (a documented dominant-label proxy, a rigorous per-label multi-label conformal analysis, and a conformal-risk-control arm), and—to test generalization beyond Europe—on the non-European, multi-continent GEO-Bench So2Sat LCZ42 benchmark (Zhu et al., 2020). Our analysis is preregistered; where the preregistered *universal*-restoration criterion was not met we preserve that outcome verbatim (§4.4) and report the conditional, encoder-dependent result as its honest reframe. Fig. 1 summarizes the audit pipeline end to end.

2. Methods

The data come from EuroSAT-MS (Helber et al., 2019) (Sentinel-2 L2A, 13 bands, 27,000 tiles, 10 classes). The shift is a *real* geographic cross-region holdout using per-tile longitude,

$$\begin{aligned} \text{source}: &\quad \text{lon} \geq P_{33} \quad \text{(E/Central Europe)}, \\ \text{target}: &\quad \text{lon} < P_{33} \quad \text{(W/Iberian–Atlantic Europe)}. \end{aligned} \tag{1}$$

with no image transform applied. Figure 2A places the P_{33} cut on the full-Europe centroid map; Figure 2B zooms the Iberia/Atlantic target against Central Europe source tiles; Figure 2C shows land-cover labels on both sides of the same cut; Figure 2D is the longitude density with the source/target counts. Patch-level examples are in Figure 3. The P_{33} cut in Eq. (1) is a single, fixed longitude-percentile boundary chosen once in advance; a post-hoc boundary-sensitivity sweep over $P_{25}/P_{40}/P_{50}$ is reported in Results (see Discussion).

This cut also induces a class-frequency (label-prior) shift between source and target that we did not adjust for in the headline logistic probe: at P_{33}, class 5 goes from 2.62% of source tiles to 17.13% of target tiles (6.54× enriched), class 1 from 15.02% to 3.16% ($\sim 4.7\times$ depleted), class 8 from 11.60% to 4.51%, and class 9 from 14.10% to 5.04% (remaining classes shift by <2.4×; target/source prior ratios in Table 1 and Fig. 4A, under the P_{33} cut of Eq. (1); Fig. 4B is the source vs. target frequency of those five extreme classes). We re-fit the probe with class-balanced sample weighting on the same cached embeddings as a control: shift ECE at α=0.05 is Fig. 4C and the count of α at which spatial-Mondrian restores coverage is Fig. 4D (Table 2).

Table 1: EuroSAT-MS per-class frequency at the P_{33} longitude cut (source = E/Central Europe, n=18,090; target = W/Iberian-Atlantic Europe, n=8,910). Classes not listed shift by <2.4×.

Class	Source %	Target %	Target/Source ratio
5 (Pasture)	2.62	17.13	6.54× (enriched)
2 (HerbaceousVegetation)	7.68	18.07	2.35× (enriched)
9 (SeaLake)	14.10	5.04	0.36× (depleted)
8 (River)	11.60	4.51	0.39× (depleted)
1 (Forest)	15.02	3.16	0.21× (∼4.7× depleted)

The frozen encoders are Prithvi-EO-2.0-300M (Szwarcman et al., 2026) (mean-pooled patch tokens, 1024-d); Clay v1.5 (Clay Foundation, 2024)

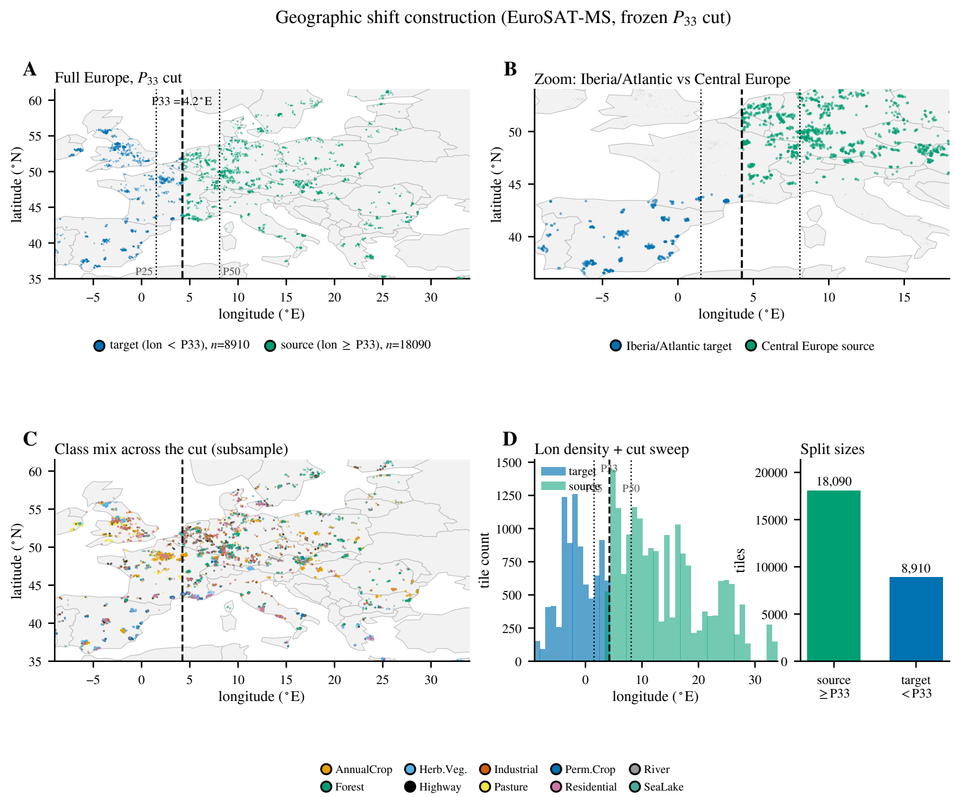

Figure 2: The geographic-shift construction on the map it is defined on (EuroSAT-MS centroids; frozen P_{33} cut = 4.2°E; source n=18,090, target n=8,910). **(A)** Full-Europe scatter (4,000/27,000 seeded subsample) colored by split side, with $P_{25}/P_{33}/P_{50}$ longitude guides. **(B)** Zoom contrasting Iberia/Atlantic target tiles against Central Europe source tiles around the cut. **(C)** Class-colored geographic mix (same cut; seeded subsample), showing land-cover labels present on both sides. **(D)** Longitude density with the same percentile guides, and the source/target tile counts. No image transform is applied — the shift is purely geographic.

(class token, 1024-d); SSL4EO-S12 (Wang et al., 2023) ViT-S/16 DINO via TorchGeo (class token, 384-d); SSL4EO-S12 (Wang et al., 2023) ViT-B/16 MAE (public SSL4EO-S12 MAE release, 87.8M params, token-free global pool, 768-d). A fifth encoder, DOFA (Dynamic One-For-All) (Xiong et al., 2024)—a ViT-B/16 whose dynamic wavelength-conditioned patch embedding ingests the Sentinel-2 bands natively (ViT-B/16 base, patch size 16; DOFA-MAE weights via TorchGeo)—was added to the five-encoder battery and probed identically on EuroSAT and on So2Sat LCZ42. The original four

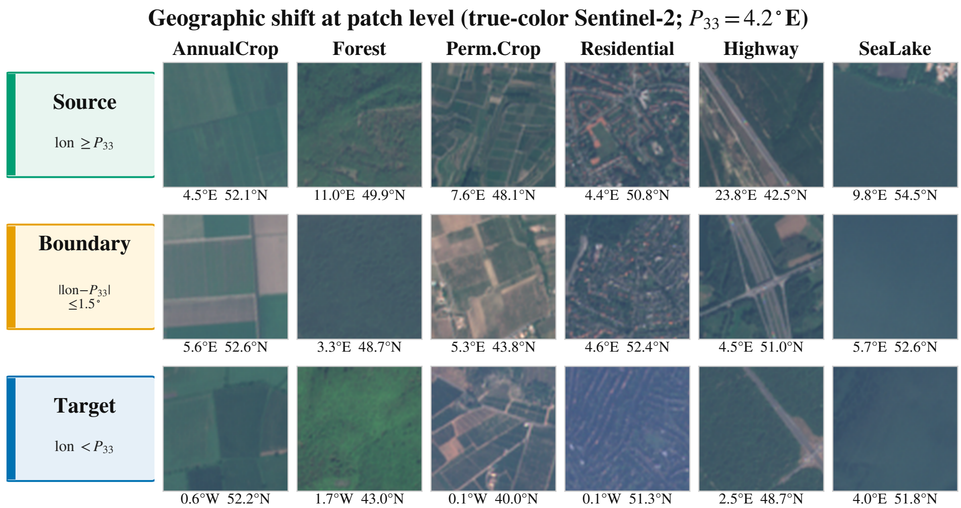

Figure 3: Geographic shift at patch level: illustrative true-color Sentinel-2 tiles (seeded selection; tile centroids annotated under each patch) for six land-cover classes (AnnualCrop, Forest, PermanentCrop, Residential, Highway, SeaLake) under the frozen P_{33} longitude cut (= 4.2°E; source n=18,090; target n=8,910). Rows encode the split sides: source (lon $\geq P_{33}$), a near-cut boundary band ($|\text{lon} - P_{33}| \leq 1.5^\circ$), and target (lon $< P_{33}$). Region-specific crop structure, settlement texture, and water color differ systematically across the cut — the shift is real but subtle at single-patch level, which is exactly why an accuracy-only deployment check can miss it.

Table 2: Class-balanced probe control (U = unadjusted headline; B = balanced; 5 seeds). (a) EuroSAT-MS: shift-ECE ranking and repair counts are unchanged. (b) BigEarthNet-S2: per-label debt, Mondrian restoration, and Mondrian set size are essentially unchanged (SSL4EO-MAE absent from part (b)).

(a) EuroSAT-MS, α=0.05

Encoder	Acc. shift (U / B)	ECE shift (U / B)	#α repaired
Prithvi	0.833 / 0.835	0.1085 / 0.1111	2/3
SSL4EO-DINO	0.880 / 0.878	0.0755 / 0.0798	1/3
SSL4EO-MAE	0.898 / 0.895	0.0650 / 0.0708	1/3
Clay	0.918 / 0.917	0.0568 / 0.0584	1/3

(b) BigEarthNet-S2, α=0.10 (per-label)

Encoder	Mean debt (U / B)	Mondrian restores	Mond. set size (U / B)
Prithvi	0.117 / 0.113	Yes	21.3 / 21.4
SSL4EO-DINO	0.239 / 0.236	Yes	20.9 / 21.1
Clay	0.163 / 0.161	Yes	20.5 / 20.6

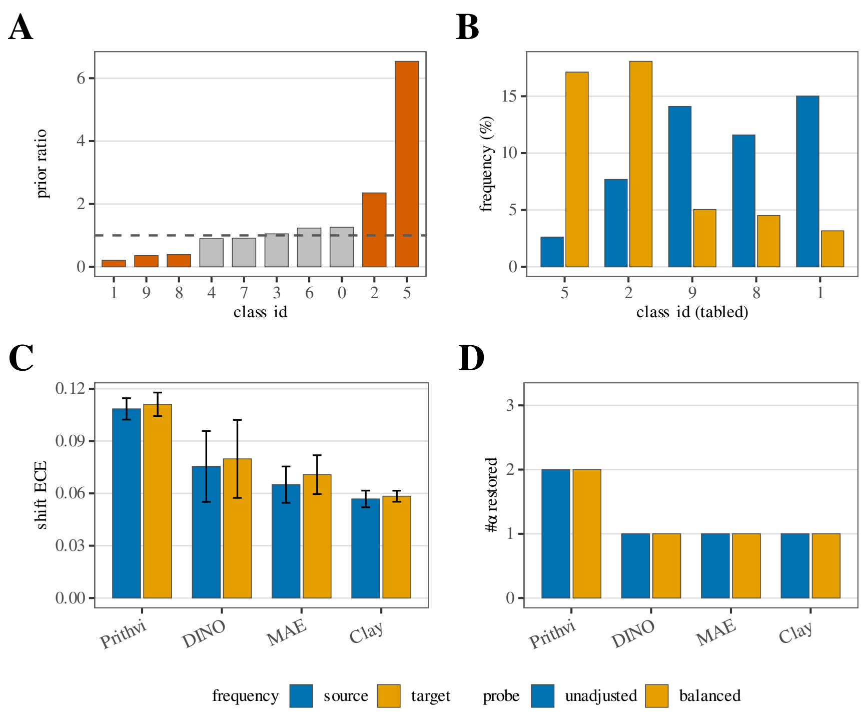

Figure 4: Class-prior / balanced-probe control at the frozen P_{33} cut (no new runs). **A:** target/source class-prior ratio by EuroSAT class id (orange = five extreme classes; grey = others); dashed line at ratio 1 (no prior shift). **B:** source vs. target class frequency (%) for those five classes. **C:** shift ECE at α=0.05 under the unadjusted (headline) vs. class-balanced probe; error bars are 5-seed t-CIs. **D:** count of $\alpha \in \{0.05, 0.10, 0.20\}$ at which spatial-Mondrian restores coverage (paired restoration CI excludes 0)—the $\#\,\alpha$ repaired count, not Mondrian coverage itself. Shared bottom legend: frequency (B) = source/target; probe (C–D) = unadjusted/balanced (same blue/orange hues, distinct meanings). Principle: the balanced probe is a *control*—encoder ranking by shift-ECE and repair counts should be stable if the P_{33} class-prior mismatch is not driving the finding.

weights load cleanly (in-distribution probe accuracy 0.95–0.99 vs. chance 0.10). For the BigEarthNet FNR analysis—the one cross-encoder comparison that turns on statistical power—we expand this battery to *thirteen* frozen encoders by adding eight more loaded through first-party published first-party weight distributions, each with normalization read from the weight's own shipped transform: SSL4EO-S12 MoCo (ViT-S/16, 384-d) and its ResNet50 MoCo/DINO variants (2048-d) (Wang et al., 2023); SoftCon (Wang et al., 2024) ViT-B/14 (768-d) and ViT-S/14 (384-d); an SSL4EO MAE ViT-L/16 (1024-d); and CROMA (Fuller et al., 2023) Base (768-d) and Large (1024-d).

Two further planned arms (a larger DOFA variant and a 600M-parameter Prithvi variant) were dropped by the pre-specified loading and mAP checks (missing or non-finite embeddings at extraction), leaving thirteen scored; the three anchor encoders reused the frozen embeddings and reproduce the earlier five-encoder records exactly. The expanded roster is used only for the BigEarthNet CRC comparison. EuroSAT retains the five-encoder battery above (Prithvi, Clay, SSL4EO-DINO, SSL4EO-MAE, DOFA). The non-European So2Sat LCZ42 extension keeps the same five-encoder *size* but substitutes SSL4EO ViT-S/16 MAE (native Sentinel-2) for SSL4EO-DINO in the ViT-S slot—so external validation is not a slot-identical checkpoint roster, and we do not claim cross-benchmark encoder ranks are interchangeable (Results).

Single-label split conformal and spatial-Mondrian recalibration.. A logistic probe is trained on frozen embeddings. Write $\hat{p}_y(x)$ for the probe's predicted probability of class y at tile x. We use the least-ambiguous-set (LAC) nonconformity (Sadinle et al., 2019)

$$s(x, y) = 1 - \hat{p}_y(x) \tag{2}$$

(higher = more nonconforming). Source-fit split conformal (Vovk et al., 2005; Angelopoulos and Bates, 2023) forms the prediction set

$$\mathcal{C}_{\text{split}}(x) = \big\{y : s(x, y) \leq \hat{q}\big\} = \big\{y : \hat{p}_y(x) \geq 1 - \hat{q}\big\}, \tag{3}$$

where $\hat{q}$ is the finite-sample $(1-\alpha)$ calibration quantile of the held-out source scores $\{s(x_i, y_i)\}$, defined with the sample size in the next display. Realized marginal coverage and mean set size are

$$\text{cov} = \frac{1}{m}\sum_{j=1}^{m} \mathbf{1}\big\{Y_j \in \mathcal{C}(X_j)\big\}, \qquad \overline{|\mathcal{C}|} = \frac{1}{m}\sum_{j=1}^{m} \big|\mathcal{C}(X_j)\big|, \tag{4}$$

with nominal level nom $= 1 - \alpha$; the set-size cost of restoring coverage under shift is reported in Results. Against this source-only baseline we run a *spatial-Mondrian* conformal predictor (Vovk et al., 2003): tiles are assigned to a fixed 4×4 spatial bin $b(x)$, and each bin draws its own quantile $\hat{q}_b$ from a disjoint same-region target-calibration slice,

$$\mathcal{C}_{\text{Mondrian}}(x) = \big\{y : s(x, y) \leq \hat{q}_{b(x)}\big\}. \tag{5}$$

We adopt the LAC score for its simplicity; adaptive prediction sets (Romano et al., 2020) are a natural alternative nonconformity we do not explore.

The Mondrian and target-calibration arms rely on exchangeability within each spatial/target bin, a weaker requirement than global exchangeability; more general non-exchangeable-shift theory (Barber et al., 2023) formalizes how conformal coverage can degrade under shifts broader than the fixed geographic partition studied here. We sweep $\alpha \in \{0.05, 0.10, 0.20\}$, 5 seeds, and bootstrap the paired per-seed restoration statistic

$$\Delta_{\text{rest}} = \left|\text{cov}_{\text{split}} - \text{nom}\right| - \left|\text{cov}_{\text{Mondrian}} - \text{nom}\right| \qquad (\text{CI}{>}0 \Rightarrow \text{restored}). \tag{6}$$

As a continuous, bucket-free debt summary we integrate the positive coverage gap over the α sweep (trapezoid rule on $\{0.05, 0.10, 0.20\}$),

$$D = \int \max\bigl(0, \text{nom} - \text{cov}_{\text{split}}(\alpha)\bigr)\, d\alpha. \tag{7}$$

A post-hoc boundary-sensitivity sweep repeats the full protocol at the P_{25}, P_{33}, P_{40}, and P_{50} longitude cuts on the same cached embeddings, and adds a third, *label-access ablation* arm: non-spatial split conformal whose single LAC quantile $\hat{q}_{\text{tgt}}$ is drawn from the same labeled target-region calibration slice the Mondrian arm uses (isolating target-label access from spatial conditioning),

$$\mathcal{C}_{\text{tgt}}(x) = \bigl\{y : s(x, y) \le \hat{q}_{\text{tgt}}\bigr\}. \tag{8}$$

Two further controls share that labeled target slice or, conversely, use no target labels. A class-conditional Mondrian arm draws a per-class quantile $\hat{q}_k$ from target-calibration examples of true class k (min-20 fallback to $\hat{q}_{\text{tgt}}$),

$$\mathcal{C}_{\text{class}}(x) = \bigl\{k : s(x, k) \le \hat{q}_k\bigr\}. \tag{9}$$

An unlabeled covariate-shift-weighted arm (Tibshirani et al., 2019) reweights source calibration scores by a density-ratio $w(x)$ (logistic source-vs-target discriminator on the frozen embeddings) and takes the weighted quantile

$$\hat{q}(x) = \min\Bigl\{s_{(m)} : \sum_{i \le m} w_{(i)} \ge (1{-}\alpha)\bigl(W + w(x)\bigr)\Bigr\}, \tag{10}$$

with $W = \sum_i w_i$ (exact split-conformal when $w \equiv 1$). A label-shift control (Podkopaev and Ramdas, 2021) instead reweights source scores by the class-prior ratio

$$w(y) = \hat{\pi}_{\text{target}}(y)/\hat{\pi}_{\text{source}}(y) \tag{11}$$

(oracle target marginal, or Black Box Shift Estimation (BBSE) from unlabeled target predictions). At small calibration budgets n, the finite-sample quantile level

$$\ell_n = \frac{\lceil (n{+}1)(1{-}\alpha) \rceil}{n} \tag{12}$$

inflates above $1-\alpha$ (Eq. (12)), which is the mechanism behind the planning-curve over-coverage reported in Results. The same curves report the fraction of coverage debt repaired at budget n,

$$\rho(n) = \frac{\mathrm{cov}_{\mathrm{tgt}}(n) - \mathrm{cov}_{\mathrm{split}}}{\mathrm{nom} - \mathrm{cov}_{\mathrm{split}}}, \tag{13}$$

defined only where the denominator is materially positive (≥ 1 pp).

Multi-label conformal and CRC.. On BigEarthNet-S2 we also run per-label binary conformal on the multi-hot labels: for each active label c, an independent probe yields a set $\mathcal{C}(x)$ and we report recall-coverage

$$\Pr\big(c \in \mathcal{C}(x) \mid y_c = 1\big) \tag{14}$$

under source split-conformal and target-side spatial-Mondrian arms. Per-label split conformal offers per-class recall calibration but no finite-sample guarantee on the example-averaged false-negative rate (FNR). We therefore add a conformal-risk-control (CRC) arm (Angelopoulos et al., 2024)—itself an instance of the distribution-free risk-controlling framework of Bates et al. (2021)—on the same frozen embeddings, splits, 4×4 Mondrian grid, and per-label probes. With threshold λ and predicted label set $\mathcal{C}_\lambda(x)$, the per-example FNR loss (monotone non-increasing in λ; bounded by 1) is

$$L_i(\lambda) = \frac{\#\{\text{positives of } i \text{ missed by } \mathcal{C}_\lambda(X_i)\}}{\#\{\text{positives of } i\}}. \tag{15}$$

The $B{=}1$ empirical risk on examples with at least one positive is

$$\widehat{R}(\lambda) = \frac{1}{n_{\mathrm{eff}}} \sum_{i:P_i \geq 1} L_i(\lambda), \tag{16}$$

where n_{eff} is the number of calibration examples with at least one positive label ($P_i = \#\{\text{positives of } i\}$). CRC selects

$$\widehat{\lambda} = \inf\Big\{\lambda : \tfrac{n_{\mathrm{eff}}}{n_{\mathrm{eff}}+1}\,\widehat{R}(\lambda) + \tfrac{1}{n_{\mathrm{eff}}+1} \leq \alpha\Big\} \tag{17}$$

(Angelopoulos $B{=}1$; exact weighted-quantile form, equivalent when each positive entry is weighted by $1/P_i$). The guarantee is expectation-level, $\mathbb{E}[L(\widehat{\lambda})] \leq \alpha$, over calibration draws; nominal FNR is therefore α itself (distinct from the coverage nominal $1-\alpha$ in Eq. (4)). CRC can certify risk α only when

$$\alpha \geq \frac{1}{n_{\mathrm{eff}}+1}; \tag{18}$$

below that floor $\widehat{\lambda} = +\infty$, the set becomes all labels, and the measured FNR collapses to zero vacuously. A target-side Mondrian-CRC arm recalibrates $\widehat{\lambda}_b$ per spatial bin on the labeled target slice; bins with fewer than 10 positives reuse a single global target λ. We report CRC FNR in-distribution, under shift, and after Mondrian-CRC repair, together with mean labels/tile set size (Eq. (4)).

Conditional coverage, calibration diagnostics, and exploratory decoupling.. Results figures are generated directly from the frozen result records by the accompanying figure-generation pipeline. Expected calibration error (ECE) is reported as a scalar probe-calibration diagnostic only (mean per-label ECE on BigEarthNet; standard multiclass ECE on EuroSAT); it is never used as a conformal score. As a post-hoc, exploratory extension, after the preregistered gate (§4.4) we added three decoupling analyses on the same frozen embeddings, computed from the same cached embeddings: (i) on BigEarthNet-S2, a per-encoder comparison of the CRC arm's shifted FNR (Eq. (15)) against multi-label accuracy (macro mAP) and calibration (mean per-label ECE) analogs on the same footing (the recomputed shifted FNR reproduces the frozen CRC record, e.g. Prithvi 0.129 vs. 0.130); (ii) on EuroSAT, per-class conditional coverage of the spatial-Mondrian-repaired target sets at matched marginal coverage (≈ 0.95–0.96), summarized by the worst-class gap versus the nominal level $1 - \alpha$

$$G_{\mathrm{worst}} = (1{-}\alpha) - \min_k \mathrm{cov}_k; \tag{19}$$

(table "Worst gap" columns always mean G_{worst}; when an encoder's repaired marginal exceeds $1 - \alpha$, prose may also quote the class's shortfall vs. that marginal, which is a different quantity) and (iii) a low-shot sweep that subsamples the EuroSAT probe-training fraction to $\{0.01, \ldots, 1.0\}$ and records split set size together with the coverage$-$accuracy gap

$$G_{\mathrm{sing}} = \mathrm{cov} - \mathrm{acc}, \tag{20}$$

mapping the singleton-degeneracy boundary (when mean set size collapses to 1, coverage coincides with accuracy and $G_{\mathrm{sing}} = 0$). Correlations across encoders are Spearman rank statistics (n=13 FNR on the expanded BigEarthNet roster, n=5 conditional coverage) and are reported as directional evidence, not proofs of independence (§4.4).

3. Results

In plain terms: every frozen encoder loses reliability under a real geographic move, that loss is repaired only with labeled target-region data, and the sections below map exactly where a conformal audit adds information beyond accuracy and where it does not. A conformal reliability audit is worth its complexity only if it tells a practitioner something accuracy and calibration do not, so we lead with the regimes in which it does and with the one cross-encoder result that is adequately powered. Primary result (next): under a multi-label false-negative-rate (FNR) risk-control guarantee, all thirteen frozen encoders incur a real geographic FNR debt that only a labeled target slice repairs, and that debt is bought back with larger prediction sets (the powered FNR-versus-set-size tradeoff); accuracy itself is only a weak, directional guide to which encoder is most robust. We then map the audit's validity boundary, a deterministic property: in the marginal single-label singleton regime the set-size axis is a restatement of accuracy, so EuroSAT at full data is a mapped negative control, not a decoupling. Next, a remedy-failure result: the textbook unlabeled covariate-shift fix does not close the debt (its weights' effective sample size collapses), so only a labeled target-region slice reliably restores the guarantee. As a conditional, single-encoder finding: on EuroSAT, restoring the *marginal* coverage guarantee hides a per-class conditional-coverage collapse that neither accuracy nor ECE predicts. Only then do we report the encoder coverage-debt ranking and its target-side recalibration repair, as a secondary, descriptive diagnostic. Every axis-decoupling analysis below is post-hoc and exploratory (added after the preregistered gate; §4.4).

Geographic FNR debt under shift and the set-size tradeoff

On GEO-Bench BigEarthNet-S2 (7,669 tiles, 39 active labels; real E→W geographic shift), a conformal-risk-control (CRC) predictor (Angelopoulos et al., 2024) bounds the example-averaged FNR of Eqs. (15)–(17)—a set-valued guarantee with *no top-1-accuracy analog by construction*. We audit *thirteen* frozen encoders here: the five-encoder battery plus eight added in the roster expansion expressly to power the cross-encoder comparison (roster in §2; two further planned arms, a larger DOFA variant and a 600M-parameter Prithvi variant, were dropped by the pre-specified loading and mAP checks). The within-encoder finding is unambiguous and requires no cross-encoder statistics: for *every one* of the thirteen encoders, source-calibrated CRC meets the target FNR in-distribution (0.051–0.056 at α=0.05) yet incurs a real

FNR debt under shift (0.130 Prithvi to 0.234 ResNet50-MoCo, i.e. 2.6–4.7× nominal across the whole roster), which a target-side Mondrian-CRC arm repays (0.002–0.036; Table 3; full per-α CRC/repair numbers appear with the BigEarthNet CRC panel below). Unlike the inert EuroSAT-singleton regime (validity-boundary result below), the repair is active here. This debt-and-repair is the reliability-relevant result: the marginal guarantee a practitioner would trust degrades under a regional move—uniformly, across every backbone we tested—and is restored only with a labeled target-region slice. This pattern is not merely descriptive: treating the thirteen preregistered per-encoder point estimates as a population, a distribution-free sign test confirms that shifted CRC FNR exceeds the nominal target in all thirteen encoders (two-sided $p \approx 2.4 \times 10^{-4}$) and that target-side Mondrian-CRC returns FNR to at or below nominal in all thirteen (same test, same p); a Wilcoxon signed-rank sensitivity check agrees exactly (one-sided $p \approx 1.2 \times 10^{-4}$ for both directions, the smallest value attainable at n=13), and every one of the thirteen per-encoder bootstrap confidence intervals excludes nominal in the claimed direction, for both the debt and the repair arm. We report the one-sided figure only as a sensitivity check: the debt-and-repair direction was not discovered in this roster run but inherited from the original five-encoder confirmatory arm, so it is defensible, but the two-sided value is what we use as the headline since it needs no such defense and is already decisive. This is a population-level summary computed post-hoc over the frozen, preregistered per-encoder estimates—the per-encoder numbers are prereg and frozen, but the population test itself is not—and it supplies a second firm, statistically-tested positive result alongside the preregistered falsification of universal restoration (§4.4). Two further qualifications bound what the test shows. First, the thirteen encoders share the same target-shift BigEarthNet split and labels, so this is a distribution-free summary over thirteen preregistered per-encoder point estimates, not an i.i.d.-sample hypothesis test: the nominal p treats the sign pattern combinatorially, and shared-target dependence means the effective number of independent draws is smaller than thirteen. Second, the repair-arm direction is what the target-calibrated Mondrian-CRC guarantee is constructed to deliver, so its 13/13 reads as a verification that the guarantee holds in practice rather than as an independent discovery. Re-aggregating to the level of independent encoder families—encoders sharing a pretraining corpus and method lineage (all six SSL4EO-S12-pretrained ViT/ResNet variants as one family; SoftCon and CROMA each one family across their two size variants; Prithvi, Clay, and DOFA as singleton families, six families in total) and requiring unanimous within-family agreement, the most conservative vote rule available—still gives a unanimous 6/6 sign test in both directions, but at a far weaker two-sided $p \approx 0.031$ (one-sided $p \approx 0.016$): the family-clustered correction survives at α=0.05, but only barely, so the thirteen per-encoder bootstrap confidence intervals (all excluding nominal in the claimed direction) remain the primary evidence, not the population p-value at either n.

The one adequately powered cross-encoder result is a risk-versus-efficiency tradeoff: FNR robustness is bought with larger prediction sets. At α=0.05, the Spearman correlation between shift FNR and set size is −0.33; at α=0.10 it is −0.66 (p=0.017). Prithvi buys the best FNR with the largest sets (15.5 labels/tile)—an operating-point choice a practitioner can act on directly, independent of which encoder they picked.

Does accuracy also tell a practitioner *which* encoder to trust? Here the evidence is weaker, and we report it as directional context rather than a claim. The FNR under shift is only weakly rank-correlated with in-distribution mAP: Spearman correlation between shift FNR and mAP is +0.32 (permutation p=0.28, n=13, α=0.05; Fig. 5) and not at all with calibration (Spearman between shift FNR and in-distribution ECE is −0.16, p=0.60). The correlation is *positive*—higher accuracy never buys better FNR robustness, and the ordering never reverses to "better encoder → better reliability"—but it does not reach significance and 48 of 78 encoder pairs (62%) invert, so we do *not* claim a strong accuracy-to-reliability inversion; at n=13 this is a weak, directional statement, not proof that accuracy is uninformative. One extreme survives the expansion and carries the practical point: Prithvi, the least accurate encoder (mAP 0.432), is the *most* FNR-robust (0.130), while the two most FNR-fragile encoders—the ResNet50 CNNs (MoCo/DINO, 0.234/0.232)—sit mid-pack on accuracy; and Prithvi vs. Clay is a near-calibration-matched dissociation (ECE 0.082 vs. 0.078, differing by 0.004, yet FNR 0.130 vs. 0.176, the worse-calibrated encoder holding the better FNR). In those first five encoders the same mAP-correlation statistic read +0.90 (asymptotic p=0.037; its *exact* permutation p was already 0.083); the roster expansion regresses it toward zero, precisely the power test the expansion was run to perform, and we report the weakened result rather than the headline it displaced (§4.4).

Conditional coverage collapse after marginal repair

After the target-side spatial-Mondrian arm restores the *marginal* EuroSAT guarantee to ≈ 0.95–0.96 for every encoder (secondary results below), we ask what that marginal scalar hides: how badly does the worst individual land-cover class still under-cover? At matched marginal coverage (α=0.05, 5 seeds;

Table 3: Primary decoupling on GEO-Bench BigEarthNet-S2 (α=0.05; 13 frozen encoders, sorted by mAP): every encoder meets the target FNR in-distribution and incurs a 2.6–4.7× FNR debt under shift that the target-side Mondrian-CRC arm repairs. "FNR shift" is the source-calibrated CRC arm's shifted FNR; "repaired" is the target-side Mondrian-CRC arm; set size is CRC labels/tile under shift. Accuracy is a weak, non-significant guide to FNR robustness: Spearman correlation of shift FNR with mAP is +0.32 (p=0.28; 48/78 inversions), with ECE is −0.16 (p=0.60), and with set size is −0.33 (n=13; −0.66, p=0.017 at α=0.10).

Encoder	mAP (in)	ECE (in)	FNR shift	FNR repaired	Set size
Prithvi	0.432	0.082	0.130	0.031	15.5
DOFA	0.459	0.057	0.169	0.021	13.0
Clay	0.502	0.078	0.176	0.022	13.3
SoftCon-ViT-B	0.525	0.033	0.174	0.031	11.9
SoftCon-ViT-S	0.525	0.020	0.213	0.036	10.7
CROMA-L	0.541	0.037	0.169	0.024	11.6
CROMA-B	0.544	0.046	0.182	0.030	12.1
ResNet50-MoCo	0.548	0.073	0.234	0.002	12.5
SSL4EO-MAE	0.550	0.037	0.170	0.030	11.8
SSL4EO-MAE (ViT-L)	0.550	0.038	0.155	0.029	11.9
ResNet50-DINO	0.551	0.055	0.232	0.014	10.8
SSL4EO-DINO	0.559	0.028	0.191	0.032	10.6
SSL4EO-MoCo	0.582	0.032	0.171	0.028	10.7

Table 4, Fig. 6A) the worst-class gap G_{worst} of Eq. (19) is scrambled relative to accuracy and ECE. Spearman correlation of the worst gap with accuracy is +0.30 (p=0.62) and with ECE is −0.30 (p=0.62); 6 of 10 pairwise inversions against each, partial correlation ≈ 0 (−0.03 for acc|ece). The smoking gun: SSL4EO-DINO is mid-pack on accuracy (0.880, 3rd of 5) and ECE (0.076, 3rd) yet silently abandons a class—one EuroSAT class is covered at only 0.850 under a 0.956 marginal (Fig. 6B)—$G_{\text{worst}} = 0.100$ vs. nominal 0.95 (Table 4), about 2× the next-worst encoder's gap—and it is the same class that under-covers in all five seeds (per-seed coverage 0.826–0.871, std 0.016; the datum is seed-stable, not a single-seed artifact). Nothing in its accuracy or calibration predicts this: DOFA, the second-least accurate encoder (0.851), has the best conditional coverage (gap 0.031), while Clay—the most accurate (0.918) and best-calibrated (ECE 0.057)—has the second-worst (gap 0.048). The effect sharpens at α=0.10, where SSL4EO-DINO's worst class sits at 0.776 under a 0.926 marginal. Which frozen encoder quietly drops a class under geographic shift is invisible to accuracy and to ECE and is surfaced only by the conditional conformal audit. Honest caveat: n=5 gives low power—all $p > 0.28$ and the weak rank correlation even flips sign across α—so the claim is "no detectable/stable dependence plus large concrete inversions,"

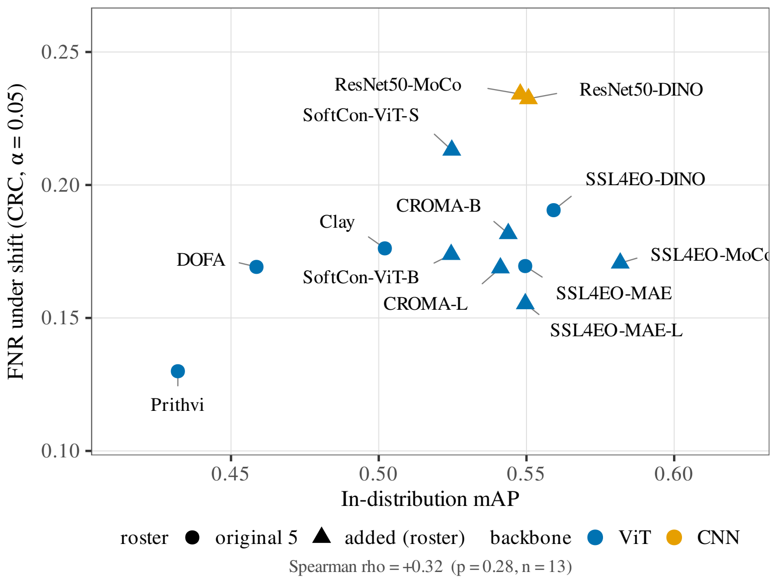

Figure 5: The five-encoder inversion does not survive expansion to thirteen. FNR under shift vs. in-distribution mAP on BigEarthNet (α=0.05, CRC arm); circles are the original five encoders, triangles the eight added at pull. The rank correlation is a weak, non-significant positive (Spearman +0.32, permutation p=0.28, n=13; no trend line is drawn, since the correlation does not reach significance): accuracy never buys better FNR robustness, but the strong +0.90 inversion of the five-encoder slice regresses toward zero. Prithvi (least accurate) remains the most FNR-robust; the two ResNet50 CNNs are the most fragile at mid accuracy.

not proven independence; the SSL4EO-DINO case is the load-bearing single datum and is reported as such (§4.4).

Singleton degeneracy and the audit validity boundary

A conformal set-size axis can decouple from accuracy only where the sets are non-trivial. Even after target-side repair the restored Mondrian set size is a deterministic image of accuracy loss (Fig. 7A): across the five encoders, Spearman correlation of set size with accuracy is -1.0 and with ECE is $+1.0$ (α=0.05, zero pairwise inversions). On EuroSAT at full data the source-calibrated split predictor emits *singletons* on the shifted target (mean set size 1.00 for every encoder and α; per-seed ≤ 1.001; Fig. 7B), so its coverage coincides with shifted top-1 accuracy (5-seed-mean $|\Delta| \leq 2 \times 10^{-4}$) and the marginal set-size/efficiency axis carries no information beyond accuracy. We therefore report set size only as a companion to a held guarantee, never as

Table 4: Per-class conditional coverage of the marginally repaired EuroSAT predictor (spatial-Mondrian, α=0.05; marginal coverage matched to $\approx$ 0.95–0.96; 5 seeds). The worst-class gap G_{worst} (coverage shortfall of the worst class vs. nominal 0.95, not vs. the encoder's marginal) does not track accuracy or ECE (Spearman $+0.30/-0.30$, p=0.62; 6/10 inversions). SSL4EO-DINO, mid-pack on both scalars, abandons a class at 0.850 (G_{worst} = 0.100; seed-stable; range 0.826–0.871); the most accurate and best-calibrated encoder (Clay) has the second-worst gap. Accuracy/ECE columns are the canonical shift values; the conditional columns are the exploratory conditional-coverage recompute.

Encoder	Acc. shift	ECE shift	Marg. cov	Worst-class cov	Worst gap
DOFA	0.851	0.093	0.951	0.919	0.031
SSL4EO-MAE	0.898	0.065	0.962	0.912	0.038
Prithvi	0.833	0.109	0.955	0.905	0.045
Clay	0.918	0.057	0.960	0.902	0.048
SSL4EO-DINO	0.880	0.076	0.956	0.850	0.100

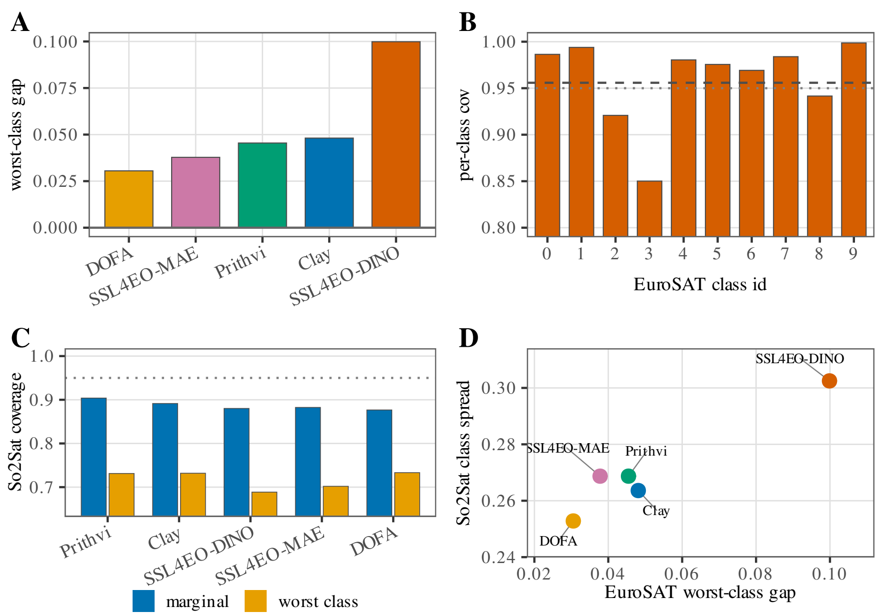

Figure 6: Conditional-coverage panels from the frozen EuroSAT and So2Sat roster summaries (no new runs; α=0.05). (A) EuroSAT worst-class coverage gap G_{worst} by encoder (Okabe–Ito colours). (B) SSL4EO-DINO per-class conditional coverage on EuroSAT (dashed = encoder marginal; dotted = nominal 0.95). (C) So2Sat source-only split coverage by encoder: marginal (blue) vs. worst class (gold); dotted = nominal 0.95. (D) Encoder scatter of EuroSAT worst-class gap vs. So2Sat class-coverage spread (same encoder colours as (A)).

evidence that the conformal layer beats accuracy—the preregistered "efficiency-decoupling" route is empirically falsified on this benchmark and we say so.

Mapping the boundary. Subsampling the probe-training fraction drives shifted accuracy down and locates where the audit switches from degenerate to informative (Table 5, Fig. 8). In the high-accuracy regime (acc $\gtrsim 0.80$ at α=0.05) sets are exact singletons (size $\equiv 1.000$) and coverage $\equiv$ accuracy (gap 0 in Fig. 8A; size $\equiv 1$ in Fig. 8B); as the task hardens below acc ≈ 0.78 the sets go non-singleton (size > 1.05) and the coverage$-$accuracy gap G_{sing} of Eq. (20) opens from 0 to +0.22 (Fig. 8A), while mean set size leaves the singleton floor (Fig. 8B). The worst-class residual under the same sweep is Fig. 8C. The more accurate the encoder, the higher the accuracy at which it degenerates (Clay singleton by acc ≈ 0.87, SSL4EO-DINO by ≈ 0.82; Fig. 8D). This makes EuroSAT at full data an explicit negative control: it sits above the boundary, so its *marginal* single-label audit is accuracy-restated—which is precisely why the per-class conditional audit above (on the non-singleton repaired sets) and the multi-label FNR audit on BigEarthNet are where the conformal layer earns its keep, and why So2Sat (17-class Local Climate Zone (LCZ), shifted accuracy 0.52–0.60), which sits below the boundary, carries informative non-singleton sets even marginally (the non-European So2Sat validation below). The boundary is itself a contribution: it tells a practitioner when a marginal conformal audit of a frozen GFM is worth running.

Coverage-debt ranking and target-side repair

The remaining EuroSAT results are the coverage-debt ranking and target-recalibration repair. They are a useful descriptive diagnostic but, per the validity-boundary result above, the marginal ranking coincides with the shifted-accuracy ranking; we report them as such, not as an independent conformal signal. Coverage debt is real and encoder-dependent: under shift, accuracy drops $0.95 \to 0.83$ (Prithvi), $0.97 \to 0.88$ (SSL4EO-DINO), $0.98 \to 0.92$ (Clay), and ECE rises to 0.109 (Prithvi), 0.076 (SSL4EO-DINO), 0.065 (SSL4EO-MAE), 0.057 (Clay)—an ordering of calibration degradation across the four encoders that is preserved under the boundary sweep and the class-prior-balanced probe control (Discussion), so it does not rest on a bare four-point monotonicity of ECE versus repair count (Fig. 9A) or of shifted accuracy versus α (Fig. 9B); the same ordering holds at spatial cuts P25–P50 (Fig. 9C). As a continuous, bucket-free debt measure we integrate the target coverage gap as in Eq. (7), $\max(0, (1-\alpha) - \text{cov}_{\text{split}})$ over the α sweep: $D = 8.0/3.3/1.4/0.8 \times 10^{-3}$ for Prithvi / SSL4EO-DINO / SSL4EO-MAE / Clay—and $D \approx 6.2 \times 10^{-3}$ for DOFA, added on the same grid as a

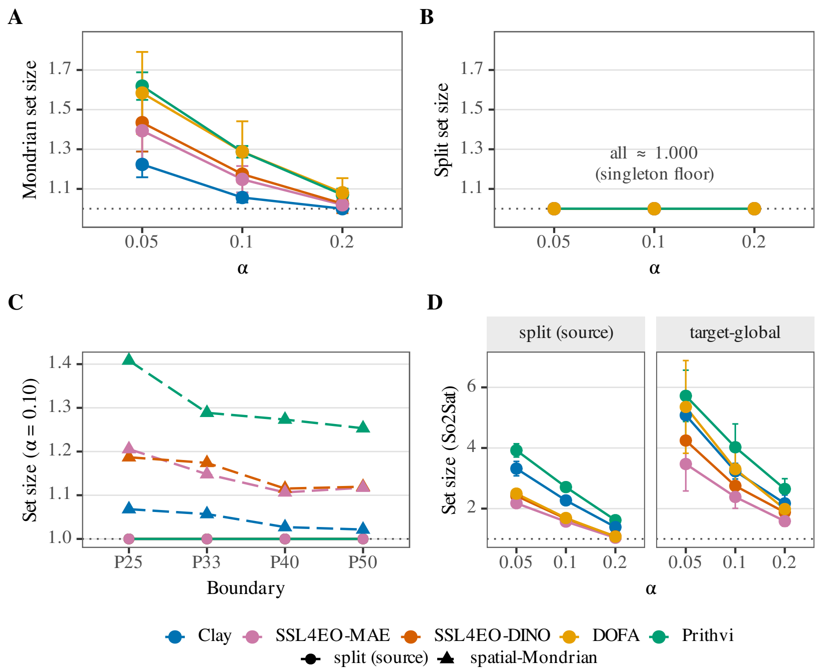

Figure 7: Four-panel set-size cost of restoration (error bars seed-level t-CIs, n=5; dotted line = singleton floor of set size 1). Shared encoder colours throughout; panel (C) uses linetype/shape for split vs. spatial-Mondrian. (A) EuroSAT spatial-Mondrian mean set size vs. α (rises above the floor only where Mondrian repays coverage debt; DOFA 1.58/1.29/1.08 at $\alpha = 0.05/0.10/0.20$). (B) Source split-conformal set size on the *same* y-scale as (A): every encoder sits on the singleton floor (≈ 1.000), so the excess size in (A) is the restoration cost. (C) Set size vs. spatial boundary percentile at α=0.10 (solid = source split; dashed = spatial-Mondrian). (D) So2Sat LCZ (17 classes) source-split vs. target-global set size in side-by-side facets—larger absolute sets than EuroSAT, on its own y-scale.

fifth encoder and the second-heaviest debt, between Prithvi and SSL4EO-DINO—recovering the same ordering without the coarse $k/3$ repaired-levels bucketing. The gap closes at $\alpha = 0.20$ for all five encoders because shifted accuracy already meets the 0.80 nominal—a no-debt artifact of the easy risk level, not evidence of a repair. DOFA's shifted split coverage is 0.851 at every risk level, so like the four battery encoders it carries under-coverage debt only at $\alpha = 0.05$ and 0.10.

Spatial-Mondrian repays the debt where it exists (restoration Δ_{rest} of Eq. (6)): at $\alpha = 0.05$ (nominal 0.95), source split-conformal under-covers (0.833 Prithvi, 0.851 DOFA, 0.880 SSL4EO-DINO, 0.898 SSL4EO-MAE,

Table 5: Singleton-degeneracy boundary (EuroSAT, α=0.05, 3 seeds; frozen low-shot probe-fraction sweep). Top: Prithvi probe-fraction sweep maps the transition from degenerate singletons (coverage $\equiv$ accuracy) to informative non-singleton sets near accuracy $\approx$ 0.78–0.80. Bottom: encoder-dependent exit from the singleton regime under the same sweep—last exact singleton (set size $\equiv$ 1) and first clearly non-singleton (set size $>$ 1.05), as shifted accuracy / train fraction. Clay remains degenerate to higher accuracy ($\approx$ 0.88) than Prithvi ($\approx$ 0.83) or SSL4EO-DINO ($\approx$ 0.82).

Train frac	Train n	Shifted acc	Split set size	Coverage − accuracy
1.00	10854	0.832	1.000	+0.000
0.50	5427	0.805	1.026	+0.011
0.25	2713	0.783	1.072	+0.026
0.10	1085	0.765	1.202	+0.066
0.05	542	0.721	1.383	+0.107
0.02	217	0.645	1.866	+0.196
0.01	108	0.650	2.294	+0.221

Encoder	Last SS$\equiv$ 1 (acc / frac)	First SS$>$ 1.05 (acc / frac)
Prithvi	0.832 / 1.00	0.783 / 0.25
Clay	0.884 / 0.10	0.818 / 0.02
SSL4EO-DINO	0.821 / 0.10	0.784 / 0.02

0.918 Clay) while spatial-Mondrian restores coverage to 0.955, 0.951, 0.956, 0.962, and 0.960 respectively, with paired-bootstrap CI excluding 0 for all five encoders (Fig. 10A). The magnitude of repair tracks the debt: at this shared risk level the restoration set size is monotone in debt—1.62 (Prithvi) $>$ 1.58 (DOFA) $>$ 1.43 (SSL4EO-DINO) $>$ 1.39 (SSL4EO-MAE) $>$ 1.22 (Clay) (Fig. 7A). At a fixed α=0.10 the Mondrian set-size cost is stable across spatial-cut percentiles (Fig. 7C). The number of risk levels repaired is weakly monotone: Prithvi (heaviest debt) and DOFA (second-heaviest) are each repaired at 2/3 levels ($\alpha = 0.05, 0.10$), while SSL4EO-DINO, SSL4EO-MAE, and Clay are each repaired at 1/3 ($\alpha = 0.05$). At $\alpha = 0.20$ (nominal 0.80) none of the five is repaired because shifted accuracy (0.83–0.92, DOFA 0.85) already meets nominal—there is no under-coverage debt to repay, and the correction correctly does nothing (Fig. 10A). The same restoration pattern holds at non-P_{33} spatial cuts (Fig. 10B). At α=0.10 the five conformal arms (split, target-global, Mondrian, weighted, class-Mondrian) are Fig. 10C; the label-shift controls (split, target-global, weighted coverage, LS-oracle, LS-BBSE) are Fig. 10D. DOFA was added to the same EuroSAT grid as a fifth encoder, with spatial-Mondrian set sizes 1.58/1.29/1.08 at $\alpha = 0.05/0.10/0.20$. It does not meet a multi-encoder universal-restoration rule on that comparison

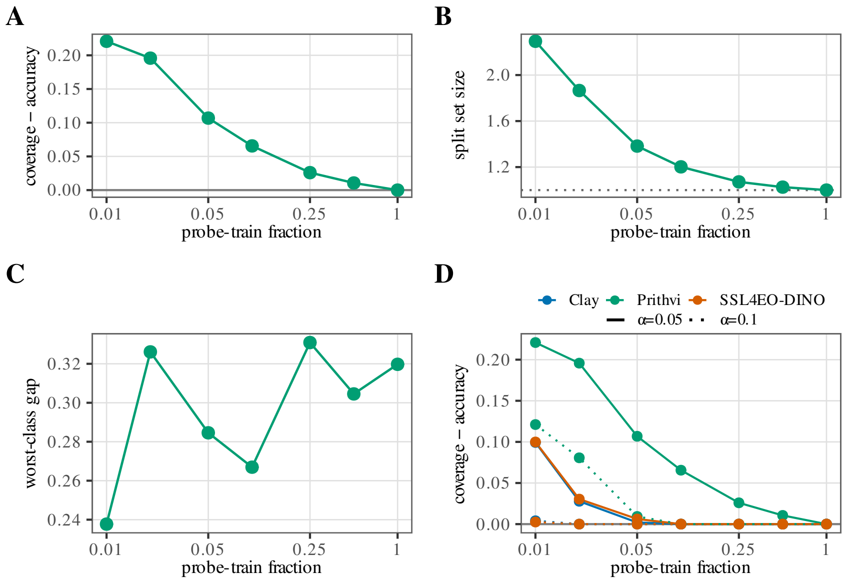

Figure 8: Singleton / low-shot validity boundary from the frozen EuroSAT low-shot sweep (no new runs). Panels (A)–(C) show the Prithvi series at α=0.05 vs. probe-train fraction (log-x). (A) Coverage−accuracy gap: as the budget rises, gap$\to$ 0 and coverage collapses onto accuracy (degenerate regime). (B) Mean split set size on the same x-axis—a *different* quantity from (A), not a restatement of the gap; the dotted line at size = 1 is the singleton floor. (C) Worst-class coverage gap vs. fraction (conditional residual under the same sweep). (D) Multi-encoder comparison (Clay, Prithvi, SSL4EO-DINO): coverage−accuracy vs. fraction with colour = encoder and linestyle solid α=0.05 / dotted α=0.10. Panels (A) and (B) are related through the singleton transition but are not redundant: gap measures coverage–accuracy decoupling, while set size measures how large the prediction sets become once that decoupling appears.

alone—Clay and Prithvi are absent from it—while per encoder the outcome is restoration at $\alpha = 0.05$ and 0.10 by the same paired-bootstrap CI-excludes-0 test applied to the four battery encoders (paired gap-difference CI low > 0), and no repair at $\alpha = 0.20$, where DOFA already over-covers (0.851). Reported DOFA numbers use the 768-d pooled embedding rather than classification-head logits. The four battery encoders are unaffected: their embeddings match the documented widths (Prithvi and Clay 1024-d, SSL4EO-DINO 384-d, SSL4EO-MAE 768-d).

Set-size cost of the same repair is in Fig. 7.

Temperature scaling (Guo et al., 2017) proves insufficient: applying the source-fit temperature to the shifted logits lowers ECE to 0.058, 0.041, 0.054,

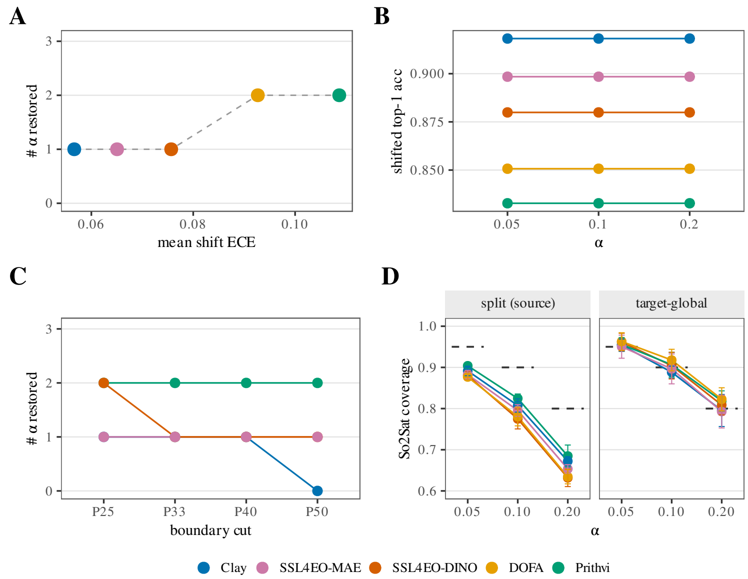

Figure 9: Four-panel coverage-debt diagnostic across five frozen encoders (shared colour legend: Clay, SSL4EO-MAE, SSL4EO-DINO, DOFA, Prithvi). (A) Mean in-shift ECE vs. the number of risk levels $\alpha \in \{0.05, 0.10, 0.20\}$ where spatial-Mondrian *restores* target coverage—i.e. the paired gap CI excludes 0 in the positive direction. ECE (the ranking diagnostic; the integrated coverage-gap D is reported in the text) orders Prithvi > DOFA > SSL4EO-DINO > SSL4EO-MAE > Clay; repair counts are weakly monotone (Prithvi = DOFA 2, others 1). (B) Shifted top-1 accuracy vs. α (flat across risk levels; ranking matches debt). (C) Boundary-sweep check on the four battery encoders (DOFA absent from that frozen sweep): $\#\alpha$ restored at spatial cuts P25–P50. (D) So2Sat LCZ cross-continent label-access check: source-split vs. target-global coverage (facets); dashed horizontal guides mark nominal $1-\alpha$ at each α (not error bars); vertical whiskers are seed-level CIs.

and 0.032 for Prithvi, SSL4EO-DINO, SSL4EO-MAE, and Clay, from the raw 0.109, 0.076, 0.065, and 0.057, but the residual still sits 2–11× above the in-distribution level (0.026, 0.014, 0.005, and 0.013), motivating the distribution-free conformal route; a focal-loss-based calibration alternative (Mukhoti et al., 2020) is likewise post-hoc-parametric and is not explored here.

The headline is not a single-cut artifact: sweeping the source/target longitude cut over $P_{25}/P_{33}/P_{40}/P_{50}$ on the same cached embeddings (5 seeds each; reported with the robustness battery below) preserves the shift-ECE encoder ordering Prithvi > SSL4EO-DINO > SSL4EO-MAE > Clay at every boundary (Prithvi 0.144/0.108/0.103/0.093 vs. Clay 0.074/0.057/0.051/0.041

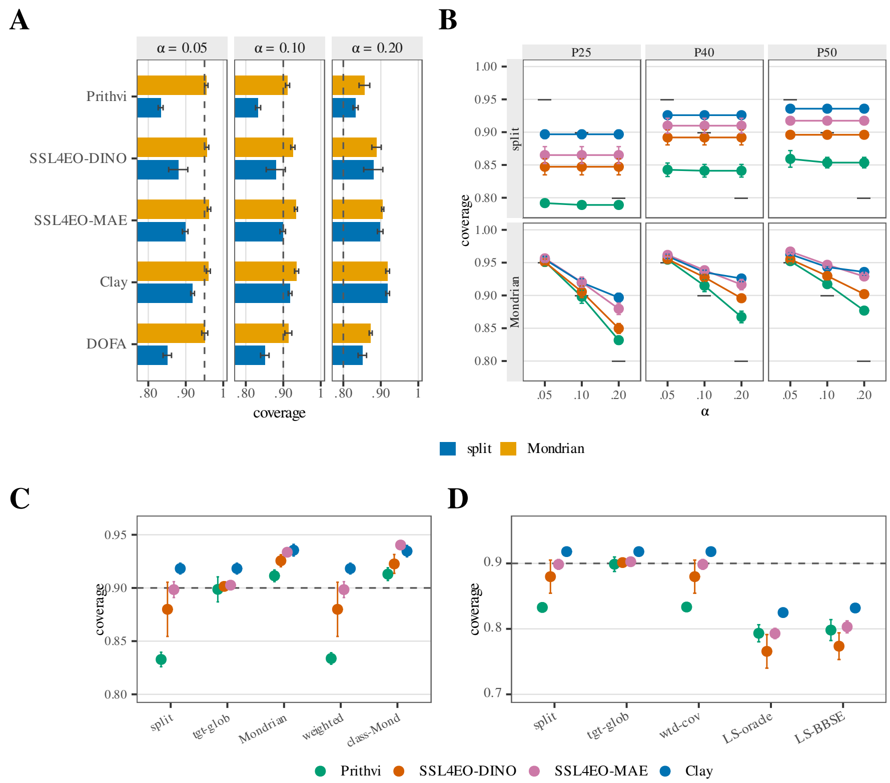

Figure 10: Coverage restoration under real geographic shift (error bars = seed-level t-CIs, n=5; dashed = nominal $1-\alpha$). (A) EuroSAT P_{33}: source split-conformal vs. spatial-Mondrian coverage by encoder and α (five encoders, including DOFA). (B) Same split vs. Mondrian contrast at non-P_{33} spatial cuts (P25/P40/P50); row = arm, colour = encoder (four-encoder battery; DOFA absent from this frozen record). (C) Five conformal arms at α=0.10 (ticks: split, tgt-glob, Mondrian, weighted, class-Mond). (D) Label-shift controls at α=0.10 (ticks: split, tgt-glob, wtd-cov, LS-oracle, LS-BBSE). Spatial-Mondrian restores where debt exists; DOFA's shifted split coverage is 0.851 at every α and spatial-Mondrian lifts it to 0.951/0.913/0.873.

at $P_{25}/P_{33}/P_{40}/P_{50}$), and the debt shrinks monotonically as the cut moves east and the target becomes less extreme (Prithvi split coverage 0.792 $\rightarrow$ 0.859). The ordering is over a small roster (five encoders at P_{33}, four across the sweep) and some adjacent gaps sit within seed noise—e.g. SSL4EO-DINO and SSL4EO-MAE differ by only $\approx$0.001 shift-ECE at P_{25}—so the extremes (Prithvi heaviest, Clay lightest) are robust while adjacent mid-rank ties should not be over-interpreted. At $\alpha = 0.05$, Mondrian restoration (paired-seed CI excluding 0) holds in 15 of 16 encoder$\times$boundary cells; the sole exception is Clay at P_{50}, where split coverage is already 0.936 (debt $\approx$0.014)—exactly the debt-targeted behaviour the conditional framing predicts. At $\alpha = 0.10$, Prithvi (heaviest debt) is restored at all four boundaries while the lighter-debt encoders mostly are not.

A label-access ablation shows that target calibration, not spatial binning, is the active ingredient: the non-spatial target-calibrated arm of Eq. (8) restores coverage to 0.946–0.952 at $\alpha = 0.05$ in every debt cell—statistically indistinguishable from spatial-Mondrian: the paired-seed "Mondrian minus target-global" gap-reduction CI excludes 0 in zero of 48 (FM$\times$boundary$\times\alpha$) cells, while target-global itself beats source-only split-conformal in the 22 cells where debt exists. Mondrian's set sizes are consistently larger (P_{33}, $\alpha = 0.05$: 1.62 vs. 1.56 Prithvi; 1.44 vs. 1.30 SSL4EO-DINO; 1.39 vs. 1.20 SSL4EO-MAE; 1.22 vs. 1.12 Clay). On this benchmark and 4×4 grid, then, the repair is bought by access to labeled target-region calibration data; per-spatial-bin conditioning adds no measurable marginal coverage benefit. We report this honestly and adjust the claim: the operative correction is *target-side conformal recalibration*, of which spatial-Mondrian is one (slightly costlier) instantiation here; finer spatial heterogeneity, where within-target variation is stronger, may still separate the two arms. Once spatial conditioning is inert, the remedy reduces to standard split-conformal calibrated on a small labeled target-region slice: we introduce no new conformal method, and spatial binning is reported here as an honest negative. The contributions of this paper are correspondingly the encoder-robustness diagnostic and the preregistered audit, not a new estimator.

How small a labeled target-region slice does the repair require? The paper prices the remedy at a single point—the fixed 30% target-calibration fraction embedded in the frozen protocol—so we sweep the budget post-hoc over 0.5%–30% of the target region on the same cached embeddings, holding the evaluation slice fixed at the frozen cut so every budget is scored on the same test examples (Fig. 11A at α=0.10; Fig. 11B is the BigEarthNet CRC arm below; Fig. 11C at α=0.05, *descriptive, not confirmatory*; Fig. 11D is $\rho(n)$ of Eq. (13)). The curve reveals a non-obvious structure: the repaired arm clears the nominal level at the smallest swept budget and then *decreases* toward nominal as the budget grows, rather than rising monotonically. This is finite-sample conformal conservatism, not a gradual repair: the finite-sample quantile level of Eq. (12) inflates well above $1-\alpha$ at small n (at n=44 samples, ℓ_n=0.932 vs. nominal 0.900 at α=0.10), over-covering and paying for the slack in larger prediction sets. As the budget grows toward 30% the inflation shrinks to $\approx 5\times10^{-4}$ and coverage converges onto nominal—from far above it at the cheapest budget (Prithvi at 0.5%: 0.994/0.959/0.904 at α=0.05/0.10/0.20) down to the frozen operating point's 0.952/0.899/0.833, which is where the paper's headline numbers sit. For a practitioner this means the labeled budget controls set-size tightness, not coverage validity: wherever debt exists, any labeled slice—down to the finite-sample floor—already repairs the guarantee, so the operating-point choice between a small, conservative slice and a larger, tighter one is a pure set-size–versus–annotation-cost tradeoff, encodable as a familiar planning curve. The exchange rate is steep and worth quoting: for Prithvi, moving from the frozen n=2673 down to n=44—a 61× annotation saving—inflates the mean set from 1.56 to 3.66 labels at α=0.05 (+134%), from 1.22 to 1.77 at α=0.10 (+46%), and from 1.00 to 1.28 at α=0.20 (+28%). The cheapest budget is therefore valid but blunt, and the 10-class α=0.20 column shows why the tightening saturates: at 30% the sets are already exact singletons (1.00), the floor beyond which no further annotation can buy anything. The reading is conditional in the same way the rest of the repair analysis is: where source-only split coverage already meets nominal there is nothing for the budget to buy. Clay at α=0.20 is the degenerate case—the curve is exactly flat at the split value 0.9180 at every budget from 0.5% to 30%—which is the singleton regime of the validity boundary (Table 5) reasserting itself: once the sets are singletons, no calibration budget can change them. Clay at α=0.10 is the intermediate case, descending from 0.9462 onto the split value rather than rising above it. The practical consequence is a deflationary one for the 30% figure the report card quotes: it is a *sufficient* budget, not a necessary one, and we relabel it as such. The BigEarthNet CRC arm shows the same structure (Fig. 11B), and adds a hard floor the EuroSAT arm does not have: CRC can only certify risk α when the attainability condition of Eq. (18) holds ($\alpha \geq 1/(n_{\text{eff}}+1)$), so at α=0.05 at least 19 calibration examples carrying a positive label are needed before any finite-sample guarantee can be issued at all. Below that the threshold diverges, the predicted set becomes every label, and the measured FNR collapses to zero *vacuously*; we exclude those budgets rather than report them as a free repair. That floor is itself the practitioner-facing number: it is a minimum labeling budget below which the guarantee is unavailable at any price. It binds in exactly one swept cell: at α=0.05 the 0.5% budget (n=12) is unattainable for all five seeds, and its apparent FNR=0.0000 is the 27.2-label average set that flags it as vacuous; at α=0.10 the same n=12 is attainable ($1/13 < 0.10$) and behaves normally. Above the floor the mirror image of the EuroSAT picture appears: FNR *rises* toward the target as the budget grows rather than falling to meet it—at α=0.05, 0.022 at 1% (n=25) against 0.051 at 30% (n=759), and at α=0.10, 0.059 at 0.5% against 0.103

at 30%—while the source-calibrated arm sits at 0.129 and 0.205. Both are the conservative side of the guarantee, and again the budget is paid for in set size: the cheapest usable budget carries 23.7 labels per set against 21.0 at 30% (α=0.05), a ≈13% inflation. The practical reading matches EuroSAT: ≈1% of the target region already discharges the risk debt, and the remaining 30× of annotation buys tighter sets, not a guarantee that would otherwise be missing.

The next three tables are a robustness battery on the EuroSAT cut, not three independent findings: each holds the frozen encoders and the coverage debt fixed and swaps exactly one variable—the correction method (unlabeled covariate-shift reweighting in Table 6; label-shift-adjusted reweighting in Table 7) or the source/target geographic cut (Table 8). BigEarthNet's multi-label FNR arm and the non-European So2Sat benchmark appear in their own subsections below. In every table the single comparison worth making is the same one: at the shared $\alpha = 0.05$ row, read the source-only (split) figure against the labeled target-side repair figure in that row—the gap between them is the debt each table is stress-testing, and it is the only number that would change the paper's claim if it closed.

A stronger unlabeled baseline still does not close the debt. Is the labeled target slice truly necessary, or would a principled unlabeled remedy suffice? As a post-hoc exploratory baseline on the cached EuroSAT features we add covariate-shift-weighted split conformal (Tibshirani et al., 2019) (Eq. (10)): density-ratio weights $w(x)$ from a logistic source-vs-target discriminator on the frozen embeddings, weighted-quantile calibration on the source scores, scored on the same shifted target (Table 6). Reweighting recovers part of the debt—at $\alpha = 0.05$ target coverage rises from source-split 0.833/0.880/0.899/0.918 to 0.919/0.938/0.911/0.927 (Prithvi/SSL4EO-DINO/SSL4EO-MAE/Clay)—but under-covers the nominal 0.95 for every encoder and never matches the labeled target-global arm (≈0.95). The mechanism is visible in the weights: the effective sample size is only 0.1–1.6% of the calibration set, i.e. the density ratios are so heavy-tailed that a handful of points dominate—a direct signature that this real E→W shift is not the pure covariate shift under which weighted conformal is guaranteed (it also carries the class-prior shift documented in Methods). The honest reading reinforces the label-access finding: even a principled unlabeled correction is insufficient and unstable here, so a small labeled target slice remains the reliable remedy.

A label-shift-adjusted control rules out the class-prior confound: the P_{33} cut's severe class-prior shift (Table 1) raises the possibility that the coverage debt above is a textbook label-shift effect (Podkopaev and Ramdas, 2021)

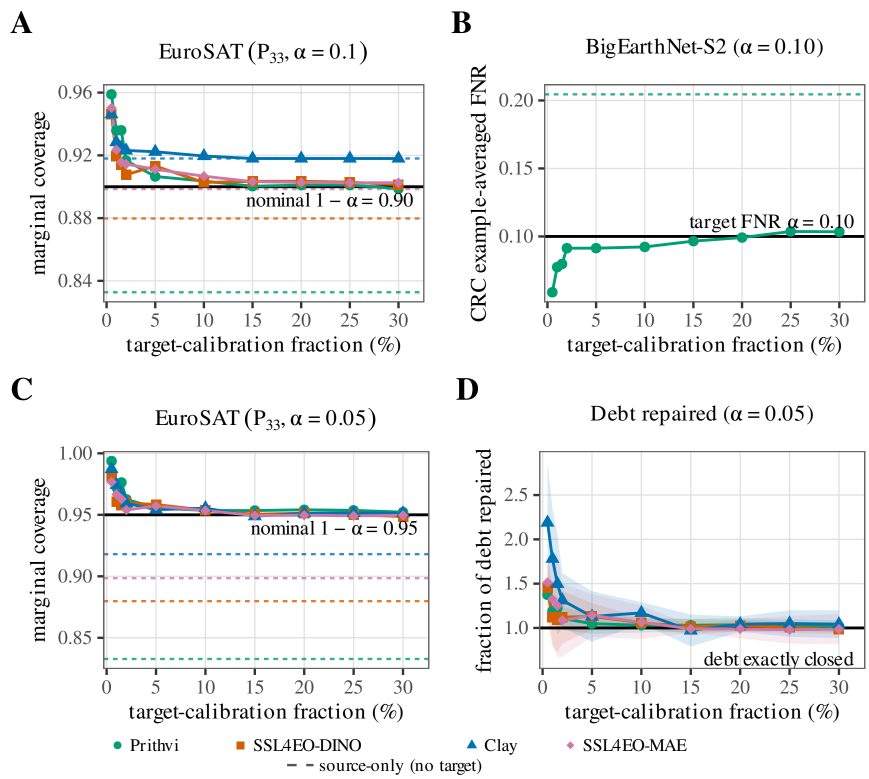

Figure 11: Target-recalibration planning curves: annotation budget versus guarantee tightness. **(A)** EuroSAT-MS (P_{33}, α=0.10): marginal coverage of the target-calibrated arm versus the labeled fraction of the target region (%). **(B)** GEO-Bench BigEarthNet-S2 (α=0.10): CRC example-averaged false-negative rate (FNR) on the same x-axis (Prithvi only; vacuous CRC budgets with $\alpha < 1/(n_{\text{eff}}+1)$ omitted). **(C)** EuroSAT-MS at the stricter α=0.05, same coverage axes as (A). **(D)** Normalized EuroSAT reading at α=0.05: fraction of coverage debt repaired versus budget (shaded bands = five-seed intervals; values >1 are over-repair). Solid coloured curves are the four encoders. Matching coloured dashed horizontals are source-only (no target labels) baselines, independent of the annotation budget. Solid black horizontals are the design targets: nominal coverage $1-\alpha$ in (A) and (C), target FNR α in (B), and exact debt clearance (= 1) in (D). The repair is not a rising ramp: wherever debt exists, the smallest usable budget already clears the nominal level, after which curves converge onto it from the conservative side. That shape is finite-sample conformal correction—the effective quantile $\lceil (n+1)(1-\alpha) \rceil / n$ inflates above $1-\alpha$ at small n (e.g. 0.932 vs. 0.900 at n=44, α=0.10), over-covering and paying for the slack in larger sets—so extra labels buy tightness to nominal, not validity of the guarantee. Descriptive secondary analysis; So2Sat excluded (no local features).

Table 6: Covariate-shift-weighted conformal as a post-hoc exploratory unlabeled baseline (EuroSAT E→W; 5-seed means; full $\alpha \in \{0.05, 0.10, 0.20\}$ sweep from the same frozen result record, expanding the α=0.05 headline cell). Target coverage under source-only split, covariate-shift-weighted split (density-ratio weights from a source-vs-target discriminator on the frozen embeddings), and the labeled target-global arm, against nominal $1-\alpha$. "ESS" is the weights' effective sample size as a fraction of the source calibration set; it does not depend on α (the weights are fixed before calibration), so it repeats per encoder. Its collapse signals that the shift is not pure covariate shift. Weighted conformal helps but under-covers nominal at every α for every encoder; only the labeled target-global arm reliably reaches it.

α	Encoder	Split cov.	Weighted cov.	Target-global	ESS frac.
0.05	Prithvi	0.833	0.919	0.952	1.6%
0.05	SSL4EO-DINO	0.880	0.938	0.949	1.3%
0.05	SSL4EO-MAE	0.899	0.911	0.950	0.4%
0.05	Clay	0.918	0.927	0.951	0.1%
0.10	Prithvi	0.833	0.881	0.899	1.6%
0.10	SSL4EO-DINO	0.880	0.914	0.901	1.3%
0.10	SSL4EO-MAE	0.899	0.905	0.903	0.4%
0.10	Clay	0.918	0.923	0.918	0.1%
0.20	Prithvi	0.833	0.855	0.833	1.6%
0.20	SSL4EO-DINO	0.880	0.898	0.880	1.3%
0.20	SSL4EO-MAE	0.899	0.902	0.899	0.4%
0.20	Clay	0.918	0.921	0.918	0.1%

rather than a genuinely geographic one. We test this directly at the identical P_{33} split: we reweight the source calibration scores by the class-prior ratio of Eq. (11) (Podkopaev–Ramdas label-shift-weighted split conformal), in an *oracle* variant (the true target-region label marginal, an upper bound on what prior-shift correction alone could repay) and a deployable variant (target priors from confusion-matrix Black-Box Shift Estimation (Lipton et al., 2018), using unlabeled target predictions only). Neither restores coverage. For three of four encoders (Clay, SSL4EO-DINO, SSL4EO-MAE) both variants make coverage worse than doing nothing (e.g. Clay 0.918 → 0.891 oracle, 0.895 BBSE at $\alpha = 0.05$; Table 7), and for Prithvi (heaviest debt) both recover only part of the gap (0.833 → 0.879/0.884) while remaining far below both nominal and the labeled target-global arm (0.952). The overcorrection compounds at looser risk levels—at $\alpha = 0.20$ every encoder's label-shift-adjusted coverage sits 8.1–17.5 pp below nominal (Clay least at ≈8 pp, SSL4EO-DINO most at ≈17.5 pp)—and more than 15 pp below even the do-nothing source-split arm. Because label-shift weighting is provably valid only when $P(x \mid y)$ is shiftinvariant, this systematic failure rules out a pure class-prior-shift explanation and is consistent with a within-class conditional shift in the frozen encoders' features across regions—the geographic mechanism this paper posits, which target-side recalibration (which assumes nothing about $P(x \mid y)$) correctly repairs and label-shift reweighting cannot.

Table 7: Label-shift-adjusted split conformal as a decisive control for class-prior confounding (EuroSAT E→W; 5-seed means; full $\alpha \in \{0.05, 0.10, 0.20\}$ sweep from the same frozen result record, expanding the α=0.05 headline cell). Oracle uses the true target-region label marginal; BBSE estimates it from unlabeled target predictions only. Neither restores coverage to nominal $1-\alpha$ at any α; three of four encoders are made worse than the do-nothing split arm, and the deficit widens at looser α (matching the $\alpha = 0.20$ pp-deficit numbers quoted above).

α	Encoder	Split	LS-oracle	LS-BBSE	Target-global
0.05	Prithvi	0.833	0.879	0.884	0.952
0.05	SSL4EO-DINO	0.880	0.858	0.867	0.949
0.05	SSL4EO-MAE	0.899	0.856	0.865	0.950
0.05	Clay	0.918	0.891	0.895	0.951
0.10	Prithvi	0.833	0.793	0.798	0.899
0.10	SSL4EO-DINO	0.880	0.766	0.774	0.901
0.10	SSL4EO-MAE	0.898	0.793	0.803	0.903
0.10	Clay	0.918	0.825	0.832	0.918
0.20	Prithvi	0.833	0.675	0.682	0.833
0.20	SSL4EO-DINO	0.880	0.625	0.639	0.880
0.20	SSL4EO-MAE	0.898	0.688	0.704	0.898
0.20	Clay	0.918	0.711	0.719	0.918

Multi-label replication on BigEarthNet-S2

We replicate the phenomenon on a second real EO benchmark: GEO-Bench (Lacoste et al., 2023) BigEarthNet-S2 (BigEarthNet-S2 (Sumbul et al., 2019); 7,669 usable tiles, real lat/lon, 36.96–67.80°N). We apply two complementary analyses.

A documented dominant-label single-label proxy (12 classes, ≥ 60 tiles each) lets the single-label LAC harness apply unchanged. All three frozen FMs replicate the pattern: ECE degrades ($0.36 \rightarrow 0.56$ Prithvi, $0.15 \rightarrow 0.42$ SSL4EO, $0.33 \rightarrow 0.59$ Clay) and spatial-Mondrian restores coverage to near-nominal at all three risk levels (Fig. 12). The proxy is a hard task (frozen-probe accuracy 0.50–0.61 in-distribution, dropping to 0.27–0.37 under

Table 8: Boundary-sensitivity sweep, real-shift target coverage "split / target-global / spatial-Mondrian" (5-seed means; split = source-only calibration; target-global = non-spatial, target-calibrated ablation arm; full $\alpha \in \{0.05, 0.10, 0.20\}$ sweep from the same frozen result record, expanding the α=0.05 headline row). † = Mondrian-vs-split paired CI does not exclude 0 (no significant restoration). At α=0.05, restoration holds in 15 of 16 encoder×boundary cells (the sole exception, Clay at P_{50}, where debt is ≈0.014, is marked); at looser α most cells no longer need repair because shifted accuracy already approaches nominal.

α	Encoder	P_{25}	P_{33} (headline)	P_{40}	P_{50}
0.05	Prithvi	0.792 / 0.946 / 0.952	0.833 / 0.952 / 0.955	0.843 / 0.950 / 0.955	0.859 / 0.949 / 0.952
0.05	SSL4EO-DINO	0.847 / 0.947 / 0.951	0.880 / 0.949 / 0.956	0.892 / 0.950 / 0.955	0.896 / 0.951 / 0.955
0.05	SSL4EO-MAE	0.865 / 0.948 / 0.957	0.898 / 0.950 / 0.962	0.910 / 0.946 / 0.962	0.917 / 0.952 / 0.967
0.05	Clay	0.897 / 0.950 / 0.955	0.918 / 0.951 / 0.960	0.926 / 0.947 / 0.960	0.936 / 0.950 / 0.963†
0.10	Prithvi	0.789 / 0.899 / 0.898	0.833 / 0.899 / 0.911	0.841 / 0.902 / 0.915	0.854 / 0.902 / 0.917
0.10	SSL4EO-DINO	0.847 / 0.897 / 0.906	0.880 / 0.901 / 0.926†	0.892 / 0.901 / 0.928†	0.896 / 0.902 / 0.930†
0.10	SSL4EO-MAE	0.865 / 0.895 / 0.920†	0.898 / 0.903 / 0.934†	0.910 / 0.910 / 0.938†	0.917 / 0.917 / 0.947†
0.10	Clay	0.897 / 0.900 / 0.920†	0.918 / 0.918 / 0.936†	0.926 / 0.926 / 0.935†	0.936 / 0.936 / 0.943†
0.20	Prithvi	0.789 / 0.802 / 0.832†	0.833 / 0.833 / 0.856†	0.841 / 0.841 / 0.867†	0.854 / 0.854 / 0.877†
0.20	SSL4EO-DINO	0.847 / 0.847 / 0.850†	0.880 / 0.880 / 0.888†	0.892 / 0.892 / 0.896†	0.896 / 0.896 / 0.902†
0.20	SSL4EO-MAE	0.865 / 0.865 / 0.880†	0.898 / 0.898 / 0.905†	0.910 / 0.910 / 0.917†	0.917 / 0.917 / 0.929†
0.20	Clay	0.897 / 0.897 / 0.897†	0.918 / 0.918 / 0.918†	0.926 / 0.926 / 0.926†	0.936 / 0.936 / 0.936†

shift), so restoration is bought at large prediction-set sizes (∼6–10 of 12 classes)—a coarse, qualitative replication.

We extend the analysis with rigorous per-class binary conformal prediction on the 43-class multi-hot labels. For each of 39 well-populated labels ($\geq$ 120 global positives; 27 active per seed after per-split filtering), we fit an independent binary probe (StandardScaler, L2-logistic, $C = 1$, 5 seeds) on source-region embeddings and calibrate per-label recall-coverage $\Pr(c \in \mathcal{C}(x) \mid y_c = 1)$ (Eq. (14)) via source-only split-conformal and target-side spatial-Mondrian arms. The coverage-debt phenomenon replicates per-label: mean recall-coverage drops from 0.92 in-distribution to 0.75 under shift ($\alpha = 0.10$; mean per-label debt 0.173), with spatial-Mondrian restoring to 0.91 (per-label paired CI excludes 0 at $\alpha \in \{0.10, 0.20\}$ for all three FMs; Fig. 13). Predicted-label set sizes remain large (20.9 of 27 active labels per tile at $\alpha = 0.10$), reflecting the geographic distributional mismatch and the recall requirement. Per-label conformal is more principled than the dominant-label proxy—it offers independent per-class recall guarantees—but both yield comparably large prediction sets under this hard pan-European shift.

This conformal-risk-control (CRC) arm, which gives a finite-sample FNR guarantee, is the substrate for the paper's primary decoupling result (Table 3); here we give its full per-α calibration and repair numbers for all five frozen encoders. Per-label split-conformal offers per-class recall calibration but no

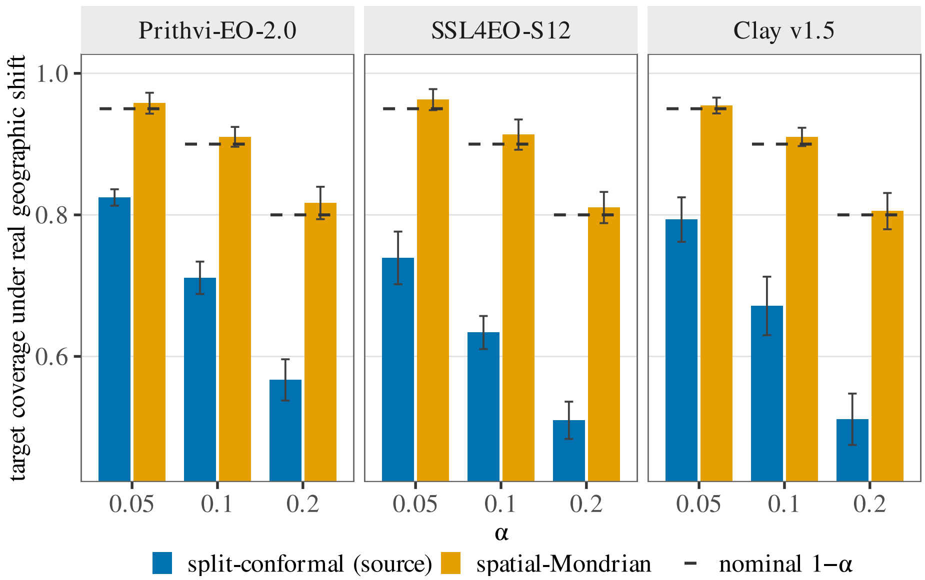

Figure 12: Dominant-label proxy on GEO-Bench BigEarthNet-S2: multi-hot BigEarthNet-S2 tiles reduced to the most frequent of 12 well-populated classes under a real geographic E→W shift (single-label LAC harness unchanged). Columns are the three frozen encoders (Prithvi-EO-2.0, SSL4EO-S12, Clay v1.5). Within each panel, blue bars are source-only split-conformal and orange bars are target-side spatial-Mondrian at $\alpha \in \{0.05, 0.10, 0.20\}$. Short black dashed segments mark the nominal coverage level $1 - \alpha$ at each α (legend: nominal $1 - \alpha$); vertical whiskers on the bars are seed-level Student t intervals (n=5), not the same mark. Coverage debt and Mondrian restoration replicate across all three encoders. This proxy is a coarse, qualitative replication; the per-label multi-label analysis below is the primary BigEarthNet evidence.

finite-sample guarantee on the example-averaged false-negative rate (FNR). We therefore add a conformal-risk-control (CRC) arm (Angelopoulos et al., 2024) on the same frozen embeddings, splits, 4×4 Mondrian grid, and per-label probes (per-example FNR loss $L_i(\lambda)$ and threshold $\widehat{\lambda}$ of Eqs. (15)–(17); exact weighted-quantile calibration; 5 seeds; percentile-bootstrap CIs, B=2000). The three-step story repeats under the marginal-FNR guarantee for every encoder (Table 9): (i) source-calibrated CRC meets the target FNR in-distribution (e.g. 0.105 at α=0.10, Prithvi; Fig. 14A, in-dist arm); (ii) the same $\widehat{\lambda}$ incurs FNR debt under the geographic shift (2–4× the nominal level, e.g. 0.206–0.277 at α=0.10; Fig. 14A, shift arm); (iii) a Mondrian-CRC arm calibrated per spatial bin on the target-region slice restores marginal FNR to at-or-below nominal (Fig. 14B), at the cost of larger label sets (∼17–18 vs. ∼8–12 labels/tile at α=0.10; Fig. 14C). The same source-calibrated FNR

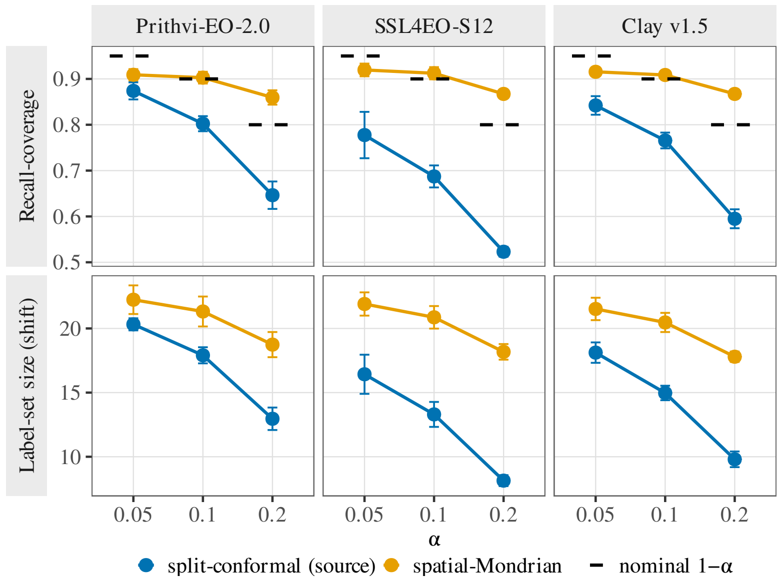

Figure 13: Per-label multi-label conformal on GEO-Bench BigEarthNet-S2 (BigEarthNet-S2; 27 active labels after per-split filtering; real geographic E→W shift). Columns are frozen encoders; rows are (top) mean per-label recall-coverage $\Pr(c \in \mathcal{C}(x) \mid y_c = 1)$ and (bottom) mean predicted label-set size per tile. Blue = source-only split-conformal; orange = target-side spatial-Mondrian; $\alpha \in \{0.05, 0.10, 0.20\}$. Short black dashed segments in the top row are nominal $1-\alpha$ guides (legend: nominal $1-\alpha$), not error bars; vertical whiskers on the points are seed-level Student t intervals (n=5). Mondrian restores per-label coverage at $\alpha \in \{0.10, 0.20\}$ (paired CI across labels excludes 0) for all three FMs. Relative to the dominant-label proxy above, this per-label analysis is primary: it supplies independent per-class recall guarantees rather than a single-label reduction.

excess under shift is present at every risk level (Fig. 14D). An honest efficiency note: CRC controls only the example-averaged FNR, so its in-distribution sets are smaller than per-label split-conformal's, and in-distribution FNR at α=0.05 fluctuates slightly above nominal (mean 0.052–0.056), consistent with the expectation-level $\mathbb{E}[L] \leq \alpha$ guarantee of Eq. (17)—which holds in expectation over calibration draws, not per individual run, so a single run landing marginally above α is expected, not a violation. Like spatial-Mondrian conformal, Mondrian-CRC requires labeled target-region calibration data—a shift-aware remedy, not a free lunch.

Non-European validation on So2Sat LCZ42

Both benchmarks above are European Sentinel-2. To test whether the coverage-debt phenomenon and its target-side repair survive outside Europe,

Table 9: CRC arm on GEO-Bench BigEarthNet-S2 (marginal per-example FNR; mean [95% percentile-bootstrap CI], B=2000, pooled over 5 seeds; $\alpha \in \{0.05, 0.10, 0.20\}$). Columns: source-calibrated CRC FNR in-distribution; the same $\widehat{\lambda}$ under the real E→W shift; target-side Mondrian-CRC FNR under shift; mean set size under shift as labels/tile of ∼27 active (CRC / Mondrian). Source-calibrated CRC tracks nominal in-distribution, incurs FNR debt under shift, and Mondrian-CRC restores it.

α	Encoder	FNR in-dist	FNR shift	Mond. FNR shift	Set size CRC/Mond.
0.05	Prithvi	0.055 [0.051, 0.058]	0.130 [0.125, 0.135]	0.031 [0.029, 0.034]	15.5/22.3
0.05	Clay	0.053 [0.049, 0.056]	0.176 [0.171, 0.182]	0.022 [0.020, 0.024]	13.3/23.8
0.05	SSL4EO-DINO	0.053 [0.050, 0.057]	0.191 [0.185, 0.196]	0.032 [0.029, 0.034]	10.6/22.3
0.05	SSL4EO-MAE	0.052 [0.048, 0.055]	0.170 [0.164, 0.175]	0.030 [0.028, 0.033]	11.8/22.1
0.05	DOFA	0.056 [0.052, 0.059]	0.169 [0.164, 0.175]	0.021 [0.019, 0.024]	13.0/24.1
0.10	Prithvi	0.105 [0.101, 0.110]	0.206 [0.200, 0.212]	0.083 [0.079, 0.086]	11.9/18.2
0.10	Clay	0.108 [0.103, 0.113]	0.277 [0.271, 0.283]	0.082 [0.078, 0.086]	9.7/18.2
0.10	SSL4EO-DINO	0.107 [0.102, 0.112]	0.277 [0.271, 0.283]	0.082 [0.078, 0.086]	7.9/17.4
0.10	SSL4EO-MAE	0.108 [0.103, 0.113]	0.262 [0.256, 0.268]	0.084 [0.080, 0.088]	8.6/17.1
0.10	DOFA	0.107 [0.103, 0.112]	0.249 [0.243, 0.256]	0.080 [0.076, 0.084]	9.9/18.1
0.20	Prithvi	0.205 [0.199, 0.211]	0.346 [0.339, 0.353]	0.184 [0.179, 0.190]	8.0/12.9
0.20	Clay	0.203 [0.197, 0.209]	0.426 [0.419, 0.433]	0.185 [0.179, 0.190]	6.4/12.9
0.20	SSL4EO-DINO	0.206 [0.200, 0.212]	0.401 [0.394, 0.408]	0.191 [0.185, 0.197]	5.3/11.1
0.20	SSL4EO-MAE	0.209 [0.202, 0.215]	0.408 [0.401, 0.415]	0.189 [0.184, 0.195]	5.5/11.3
0.20	DOFA	0.203 [0.196, 0.209]	0.384 [0.377, 0.391]	0.187 [0.181, 0.193]	6.7/12.4

we repeat the harness on GEO-Bench (Lacoste et al., 2023) So2Sat LCZ42 (Zhu et al., 2020); 42 cities across multiple continents; 17 local-climate-zone classes), using GEO-Bench's own train/valid (source) vs. held-out test-city (target) partition as the shift. Because this release ships no per-tile latitude/longitude with the release, there is no spatial-Mondrian arm here; we compare only source-only split-conformal against a non-spatial, target-calibrated arm (*target-global*), i.e. exactly the label-access ablation that Table 8 isolated as the active ingredient on EuroSAT. All five frozen-encoder slots complete on this benchmark (Prithvi, Clay, SSL4EO ViT-S/16 MAE native-S2, SSL4EO-MAE ViT-B/16, DOFA)—with SSL4EO ViT-S/16 MAE substituted for the EuroSAT SSL4EO-DINO arm in the ViT-S slot (Methods), so this is a matched five-encoder *size* rather than a slot-identical battery—and the encoder-ranking diagnostic is available here as well.

The debt-and-repair pattern replicates outside Europe: under the held-out-city shift, accuracy drops by ≈0.14–0.18 across the five encoders (0.670 → 0.522 Prithvi up to 0.796 → 0.637 SSL4EO-MAE)—a genuine, non-trivial distribution shift. Source-only split-conformal under-covers at every risk level and every encoder ($\alpha = 0.05$: 0.877–0.904; $\alpha = 0.20$: 0.632–0.685), and the

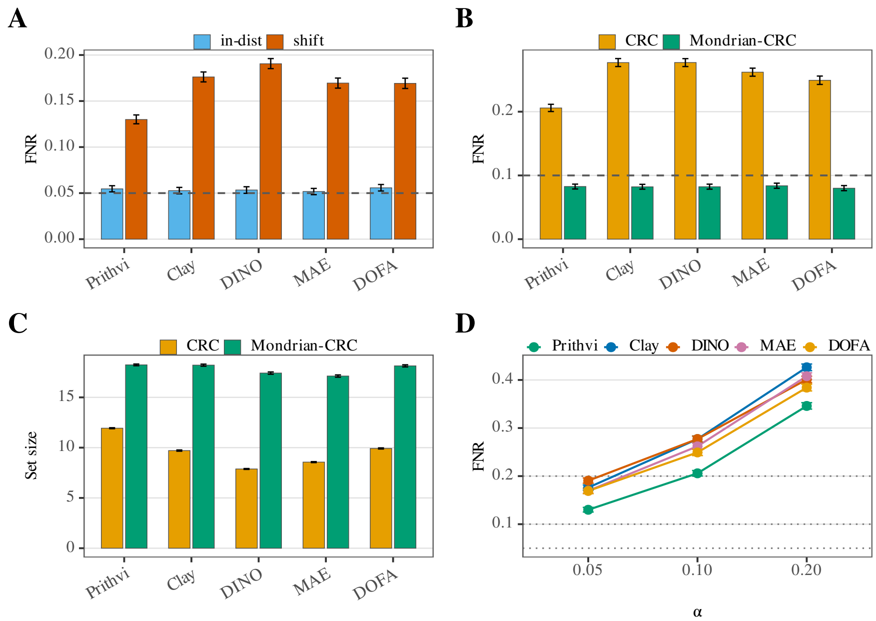

Figure 14: CRC on BigEarthNet (frozen multi-seed bootstrap means with 95% CIs; no new runs). All panels use example-averaged FNR (or set size) pooled over five seeds; error bars are percentile bootstrap 95% intervals (B=2000). Encoder ticks and the (D) colour key abbreviate SSL4EO-DINO/MAE as DINO/MAE (Prithvi, Clay, DINO, MAE, DOFA). The x-axis in (A)–(C) is the encoder; only (D) uses α as the x-axis. (A) Source-calibrated CRC FNR in-distribution vs. under the E$\to$W geographic shift at α=0.05 (blue = in-dist, vermillion = shift; dashed = nominal FNR level α). In-distribution meets the risk level; the same $\widehat{\lambda}$ incurs large FNR debt under shift. (B) Shift FNR at α=0.10: source-calibrated CRC vs. target-side Mondrian-CRC (orange = CRC, green = Mondrian-CRC; dashed = nominal α). Mondrian-CRC restores FNR to at-or-below nominal for every encoder. (C) Mean set-size cost under shift at the same α=0.10 (labels/tile of $\sim$27 active): Mondrian-CRC's restore comes with larger sets than source-calibrated CRC (shared orange/green method colours with (B)). (D) Source-calibrated CRC FNR under shift across $\alpha \in \{0.05, 0.10, 0.20\}$ by encoder (coloured lines; dotted horizontals = nominal α at each risk level).

non-spatial target-calibrated arm restores coverage to near-nominal ($\alpha = 0.05$: 0.950–0.963), with the paired-seed restoration CI excluding 0 in every one of the fifteen debt cells (5 encoders $\times$ 3 risk levels; Table 10). Restoration is bought at modest set sizes ($\sim$2–6 of 17 classes), not vacuous full-label sets. This external label-access replication now spans all five encoders on a non-European, multi-continent benchmark: it confirms the operative part of the finding—*target-side conformal recalibration*, not spatial binning, repays the debt. It still lacks a spatial-Mondrian arm (no per-tile lat/lon), so it tests the label-access mechanism only; but with all five encoders it also carries the per-class conditional-coverage diagnostic (below). A data-provenance note applies: the So2Sat LCZ42 split used here was obtained from the public GEO-Bench 1.0 release after the v0.9.1 archive was unavailable, with labels taken from per-sample ground truth (all 21,964 samples); version differences between v0.9.1 and v1.0 were not independently audited.

Table 10: Non-European external validation on GEO-Bench So2Sat LCZ42 (17 local-climate-zone classes, 42 multi-continent cities; 5-seed means). Target coverage "split / target-global" under the held-out-city shift (split = source-only calibration; target-global = non-spatial, target-calibrated label-access arm; no spatial-Mondrian arm, as this release carries no per-tile lat/lon). Set sizes (of 17 classes) appear in brackets.

α	Encoder	Split cov. [set]	Target-global cov. [set]	Acc. in→shift
0.05	Prithvi	0.904 [3.92]	0.956 [5.72]	0.670→0.522
0.05	Clay	0.891 [3.32]	0.954 [5.08]	0.713→0.569
0.05	SSL4EO ViT-S/16 MAE	0.880 [2.42]	0.962 [4.25]	0.781→0.601
0.05	SSL4EO-MAE ViT-B/16	0.882 [2.18]	0.950 [3.47]	0.796→0.637
0.05	DOFA	0.877 [2.49]	0.963 [5.35]	0.777→0.608
0.10	Prithvi	0.825 [2.71]	0.907 [4.02]	0.670→0.522
0.10	Clay	0.808 [2.27]	0.888 [3.24]	0.713→0.569
0.10	SSL4EO ViT-S/16 MAE	0.775 [1.67]	0.904 [2.75]	0.781→0.601
0.10	SSL4EO-MAE ViT-B/16	0.795 [1.57]	0.896 [2.38]	0.796→0.637
0.10	DOFA	0.781 [1.69]	0.918 [3.30]	0.777→0.608
0.20	Prithvi	0.685 [1.62]	0.819 [2.65]	0.670→0.522
0.20	Clay	0.672 [1.39]	0.795 [2.17]	0.713→0.569
0.20	SSL4EO ViT-S/16 MAE	0.632 [1.08]	0.809 [1.88]	0.781→0.601
0.20	SSL4EO-MAE ViT-B/16	0.654 [1.04]	0.793 [1.59]	0.796→0.637
0.20	DOFA	0.634 [1.07]	0.823 [1.99]	0.777→0.608

As a second, multiclass conditional-coverage dissociation, with all five encoders, So2Sat LCZ42 also lets us re-run the per-class conditional-coverage diagnostic of the EuroSAT conditional finding (§3) on a second, 17-class multiclass benchmark. At α=0.05 we measure, for the source-only split predictor, the worst individual LCZ class coverage per encoder (Table 11; Fig. 6C). It dissociates from both shifted accuracy and shift-ECE: Spearman correlation of worst-class coverage with accuracy is −0.10 (p=0.87, 5 pairwise inversions) and with shift ECE is +0.60 (p=0.29, 3 inversions); the class-coverage spread scrambles likewise ($\rho = -0.30$ against each, 6 inversions; Fig. 6D). Non-singleton set sizes on this 17-class task are Fig. 7D; source-split vs. target-global coverage is Fig. 9D. The most accurate encoder here (SSL4EO-MAE, acc 0.637) is not the one that best protects its worst class, and the worst-covered class (0.689–0.733 across encoders, under marginal coverage 0.877–0.904) is under-served regardless of accuracy or calibration ranking. This corroborates the EuroSAT conditional-coverage finding with an independent multiclass benchmark—though on the unrepaired split sets here (no target-side repair or spatial arm on this release), and, like the EuroSAT case, it is post-hoc, exploratory, and small-n (n=5, all $p > 0.28$; §4.4).

Table 11: Per-class conditional coverage on GEO-Bench So2Sat LCZ42 (17 LCZ classes; source-only split-conformal, α=0.05; 5 seeds). The worst individual-class coverage dissociates from shifted accuracy and shift-ECE (Spearman −0.10 vs. accuracy, +0.60 vs. ECE; 5/3 pairwise inversions), a second multiclass instance of the EuroSAT conditional-coverage collapse.

Encoder	Acc. shift	ECE shift	Marg. cov	Worst-class cov	Class spread
Prithvi	0.522	0.364	0.904	0.731	0.269
Clay	0.569	0.325	0.891	0.732	0.264
SSL4EO ViT-S/16 MAE	0.601	0.114	0.880	0.689	0.302
SSL4EO-MAE ViT-B/16	0.637	0.173	0.882	0.702	0.269
DOFA	0.608	0.214	0.877	0.733	0.253

The four report-card numbers previewed in the Abstract and Fig. 1 are assembled for one median-accuracy encoder in Table 12 (§4.3).

4. Discussion

4.1. Interpretation

A conformal audit of a frozen optical GFM earns its keep where accuracy and calibration go silent (Guo et al., 2017; Ovadia et al., 2019). On multi-label FNR risk control, every one of thirteen frozen encoders incurs a cross-European FNR debt under shift (2.6–4.7× nominal) that only a labeled target slice repairs; accuracy does not reliably identify the most robust encoder—the least accurate is the most robust, yet at n=13 the rank correlation is only a weak positive ($\rho = +0.32$; Table 3). On per-class conditional coverage, restoring the marginal guarantee can still leave one land-cover class covered at 0.850 under a 0.956 marginal (Table 4), again unpredictable from the scalars a practitioner would rank on. Symmetrically, the audit adds little in the high-accuracy singleton regime: marginal set size is then a deterministic restatement of accuracy, so EuroSAT at full data is a mapped negative control rather than a decoupling, and the preregistered efficiency-decoupling route is falsified. Within that boundary, the descriptive coverage-debt ranking remains useful—Prithvi > DOFA > SSL4EO-DINO > SSL4EO-MAE > Clay on the five-encoder EuroSAT debt story (extremes Prithvi/Clay; DOFA second-heaviest)—but we avoid a "universal fix" claim: where shifted accuracy already meets the nominal level, there is no debt and no repair. Relative to accuracy-centric GFM surveys and benchmarks (Xiao et al., 2025; Lacoste et al., 2023; Ghamisi et al., 2025), the increment is pricing geographic coverage debt rather than OOD accuracy alone.

4.2. Mechanisms and controls

Headline coverage and ranking numbers use an unadjusted logistic probe and therefore inherit the P_{33} source/target class-prior mismatch (Methods; up to 6.54× enrichment / ∼4.7× depletion for individual EuroSAT classes). Refitting every probe with class-balanced sample weighting on the identical cached embeddings leaves the shift-ECE ranking, the per-encoder count of risk levels repaired under the spatial-Mondrian test of Eq. (6), and the analogous per-label-balanced BigEarthNet check unchanged to within seed noise (Tables 1, 2; Fig. 4A the prior ratios, Fig. 4B the five-class frequencies, Fig. 4C the balanced vs. unadjusted ECE, Fig. 4D the Mondrian repair-count control). The prior mismatch therefore does not appear to drive the ranking; the balanced control was run only at the frozen P_{33} boundary (Table 8 remains unadjusted). The ViT-B backbone (SSL4EO-MAE) incurs less debt than the ViT-S backbone (SSL4EO-DINO), but those checkpoints also differ in objective and release, so the pairwise ordering is consistent with—yet does not isolate—a backbone-capacity mechanism. EO conformal applications establish distribution-free sets for EO pipelines (Singh et al., 2024; Lou et al., 2025; Mao et al., 2024) but do not quantify region-conditional debt for interchangeable frozen GFMs; we reuse split and spatial-Mondrian conformal prediction with CRC (Vovk et al., 2005, 2003; Angelopoulos et al., 2024) under a frozen-probe constraint rather than proposing a new estimator.

4.3. Deployment implications

Operationally, the report card is a pre-deployment gate. If the task is multi-label (or otherwise below the singleton-degeneracy boundary), treat in-distribution CRC/Mondrian coverage as necessary but not sufficient: budget a labeled target-region calibration slice before claiming a reliability guarantee, because unlabeled covariate-shift weighting collapses on effective sample size here and label-shift reweighting does not substitute (Tibshirani et al., 2019). Prefer encoder selection that jointly inspects set-size / FNR robustness and worst-class conditional coverage, not accuracy or ECE alone (Romano et al., 2020; Cauchois et al., 2021). If the task is already in the high-accuracy singleton regime, do not expect the marginal conformal layer to reveal a new ranking. Table 12 illustrates the gate for the median-accuracy encoder (Clay v1.5; mAP 0.502, rank 3 of 5) at α=0.05: uncalibrated example-averaged FNR misses the 0.05 target by $\approx 3.5\times$ (seed-mean 0.175, bootstrap 0.176); a 30% labeled target slice with Mondrian-CRC restores FNR to 0.022 at the cost of larger sets (13.3 $\rightarrow$ 23.8 of 27 labels). On So2Sat LCZ42 (LCZ-17), a healthy marginal (0.891) still hides a worst class at 0.732. Deploy only with labeled target recalibration, and monitor per-class coverage rather than the marginal alone. Relative to geographic-robustness testbeds that score OOD accuracy rather than coverage debt (Doerksen and Kerner, 2026; Al-Emadi et al., 2025; Meyer and Pebesma, 2021), the practical increment is explicit pricing of that labeled target slice.

Table 12: Deployment reliability report card (illustration; post-hoc) for Clay v1.5 at α=0.05. Multi-label FNR: GEO-Bench BigEarthNet-S2 ($E \rightarrow W$); worst-class: So2Sat LCZ42 LCZ-17 (split). Brackets: seed-level t-CIs (n=5).

Report-card question (α=0.05, encoder Clay)	Value
Multi-label FNR if deployed uncalibrated	0.175 $[0.143, 0.208]$
(relative to the 0.05 target)	$\approx 3.5\times$ over
FNR after a 30% target slice (Mondrian-CRC)	0.022 $[0.015, 0.029]$
Efficiency cost (mean set size, of 27 labels)	13.3 $\rightarrow$ 23.8
Worst class, 2nd region (So2Sat LCZ42 LCZ)	0.732 $[0.692, 0.772]$
(under a marginal coverage of)	0.891

4.4. Scope and limitations

The single preregistered confirmatory result is a falsification: Stage 2 required spatial-Mondrian restoration at ≥ 2 of 3 risk levels for both Prithvi and Clay; Clay restored at only 1/3 (Prithvi 2/3). The encoder-dependent, debt-targeted account used throughout is therefore an exploratory reframe; we introduce no new estimator, only the report-card protocol (§4.3) and the audited evidence behind it (Vovk et al., 2005; Angelopoulos et al., 2024). Beside that negative, a post-hoc sign test over the thirteen frozen per-encoder FNR estimates confirms the within-encoder debt-and-repair pattern at $p \approx 2.4 \times 10^{-4}$ in both directions (§3).

The three re-centered decoupling axes are post-hoc and, for the conditional analyses, small-n. Roster expansion weakened the cross-encoder FNR claim: five-encoder $\rho = +0.90$ (exact permutation p=0.083) regresses to +0.32 at n=13 (permutation p=0.28; 48/78 pairwise inversions). We keep only the directional claim that accuracy is not a reliable guide—the extreme and the near-calibration-matched Prithvi/Clay pair still dissociate—plus the surviving FNR-vs-set-size tradeoff ($\rho = -0.66$, p=0.017 at α=0.10). Conditional-coverage dissociation (n=5, all $p > 0.28$) supports "no detectable dependence," not proven independence.

Scope is optical Sentinel-2 with a EuroSAT-MS longitude cut, BigEarthNet replication, and one non-European LCZ extension (So2Sat LCZ42)—not continent-scale spatial-Mondrian generalization, not SAR/optical cross-sensor transfer (a preliminary S1 probe failed in-distribution control and is withheld), and not a 2025-generation leaderboard (Tseng et al., 2025; Danish et al., 2025). Covariate-shift-weighted and label-shift conformal controls fail to restore coverage (Tables 6, 7); class-conditional Mondrian remains unexecuted. The planning curve (Fig. 11) and boundary sweep (Table 8) are post-hoc extensions of the frozen 30% / P_{33} protocol. Honest follow-ons include expanding the frozen roster and geographic axes, testing whether audited or adaptive remedies (Zhou et al., 2026; Gibbs and Candès, 2021, 2024) reduce the labeled target budget, and pairing the report card with OOD flags and area-of-applicability maps (Gonzalez-Calabuig et al., 2025; Meyer and Pebesma, 2021; Tamazyan et al., 2025).

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Declaration of generative AI use

Generative AI tools were used only for auxiliary assistance. The author reviewed and verified the final manuscript and takes full responsibility for its content.

Data availability

Public datasets: EuroSAT-MS, BigEarthNet, and So2Sat LCZ42. Public foundation-model weights include Prithvi-EO-2.0-300M, Clay v1.5, SSL4EO-S12, and DOFA. Code repository: `https://github.com/PeterPonyu/geospatial-fm-reliability-research`. Archive: Zenodo DOI `10.5281/zenodo.21130299`.

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